```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[HN_FFCutsFits,CutFitFileHN] =
recover_selected_cuts('HN_FFCutsFits','e:\SHARE\Fe\Data\analysis\07_fit_with_multic
ut_allJ\sel_cuts');
```

```
fn = fieldnames(HN_FFCutsFits);
nf = 11
```

```
nf =
11
```

```
%for i = 11:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei800_offH2H0_HNminFF'
```

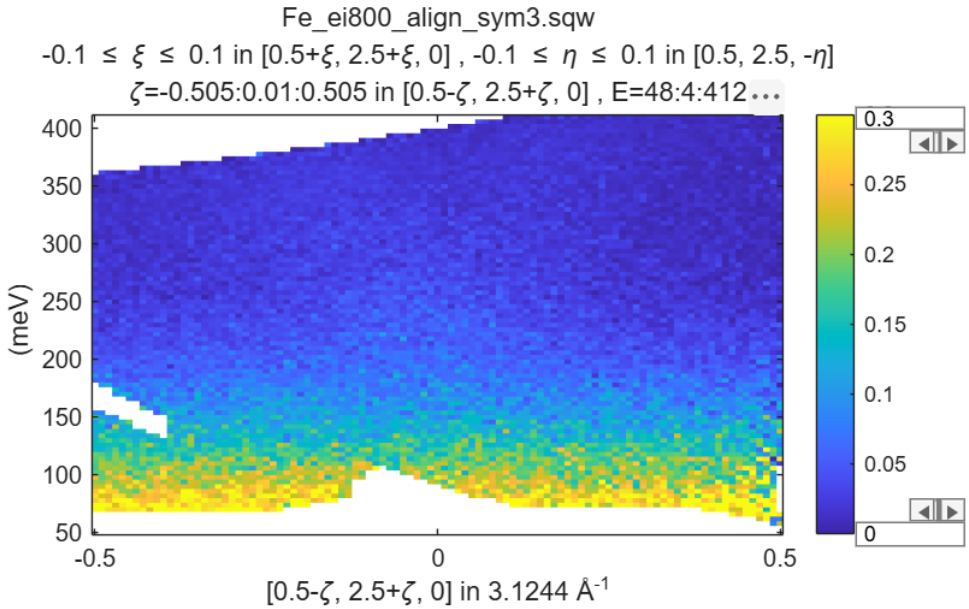
```
selectCuts =
'Ei800_offH2H0_HNminFF'
```

```
if ismember(selectCuts(1:3),{'msk','dat'})||...
        contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

fprintf('ProcessiHN cuts: %s \n', selectCuts);
```

```
ProcessiHN cuts: Ei800_offH2H0_HNminFF
```

```
cuts_list = HN_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)
    cuts_list{j} = cuts_list{j}.set_instrument(mi);
    cuts_list{j} = cuts_list{j}.set_sample(sample);
    plot(cuts_list{j}); lz 0 0.3; keep_figure;
end
```




```
dir_name  = "HN";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(min_Ei,en_mm(1));
en_mm(1) = 200;
%en_mm(2) = 380;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
HN_FFCutsFits.(selectCuts) = cuts_list;
```

```
HN_FFCutsFits = fit_cut2JFF_along_direction(...
    HN_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

```
*****************************
**** En = 205±5 going to 405
*****************************
--------------------------------------------------------------------------------
Beginning fit (max 20 iterations)----------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 2.8068s    Reduced Chi^2 = 1.4638
 Free parameter values:
           80         0.7917             39             35        0.05131
--------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 7.602s    Reduced Chi^2 = 0.73727      Levenberg-Marquardt = 0.1
 Free parameter values:
        146.8          1.355          39.14             35        0.04156
--------------------------------------------------------------------------------
Iteration = 2
------------------
```

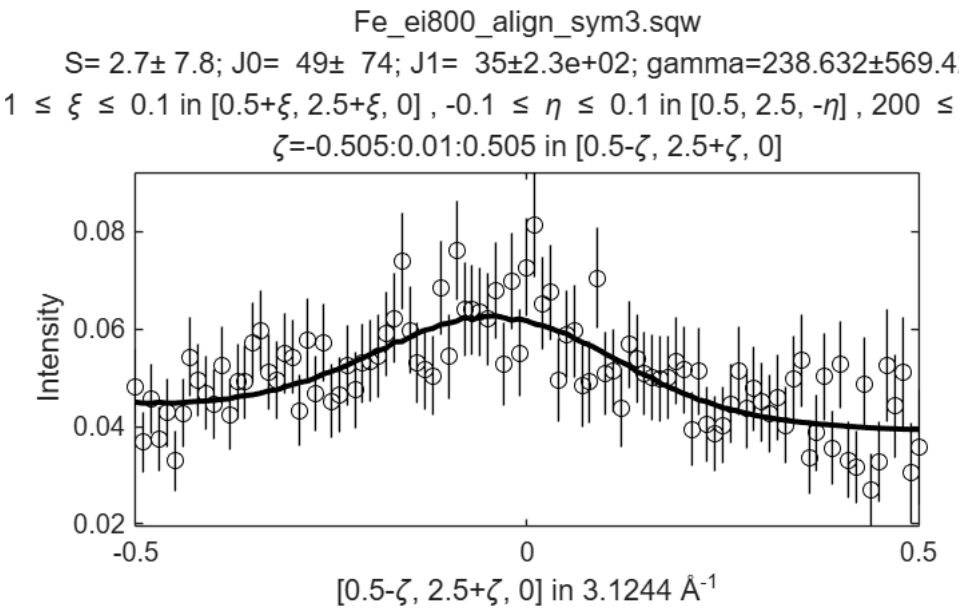
```
 Total time = 10.6181s    Reduced Chi^2 = 0.70635      Levenberg-Marquardt = 0.01
 Free parameter values:
        198.9          2.161          44.85             35        0.03854
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 13.6282s    Reduced Chi^2 = 0.70904      Levenberg-Marquardt = 0.001
 Free parameter values:
        308.3          3.704           59.2             35        0.03811
 Total time = 14.2379s    Reduced Chi^2 = 0.69869      Levenberg-Marquardt = 0.01
 Free parameter values:
        238.6          2.742          49.42             35        0.03827
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 17.2165s    Reduced Chi^2 = 0.70127      Levenberg-Marquardt = 0.001
 Free parameter values:
        335.1          4.025          61.34             35        0.03809
 Total time = 17.9443s    Reduced Chi^2 = 0.69931      Levenberg-Marquardt = 0.01
 Free parameter values:
        273.4          3.181          53.33             35        0.03819
 Total time = 18.7027s    Reduced Chi^2 = 0.70198      Levenberg-Marquardt = 0.1
 Free parameter values:
        251.7           2.89          50.59             35        0.03824
 Total time = 19.3002s    Reduced Chi^2 = 0.69999      Levenberg-Marquardt = 10
 Free parameter values:
        240.8          2.761           49.5             35        0.03828
 Total time = 19.9049s    Reduced Chi^2 = 0.69946      Levenberg-Marquardt = 1000
 Free parameter values:
        238.9          2.744          49.43             35        0.03827
 Total time = 20.5173s    Reduced Chi^2 = 0.69944      Levenberg-Marquardt = 100000
 Free parameter values:
        238.7          2.742          49.42             35        0.03827
 *** No improvement in chi-squared over previous iteration ***
 Total time = 20.5274s    Reduced Chi^2 = 0.69869
 Free parameter values:
        238.6          2.742          49.42             35        0.03827
-------------------------------------------------------------------------------
Fit converged:

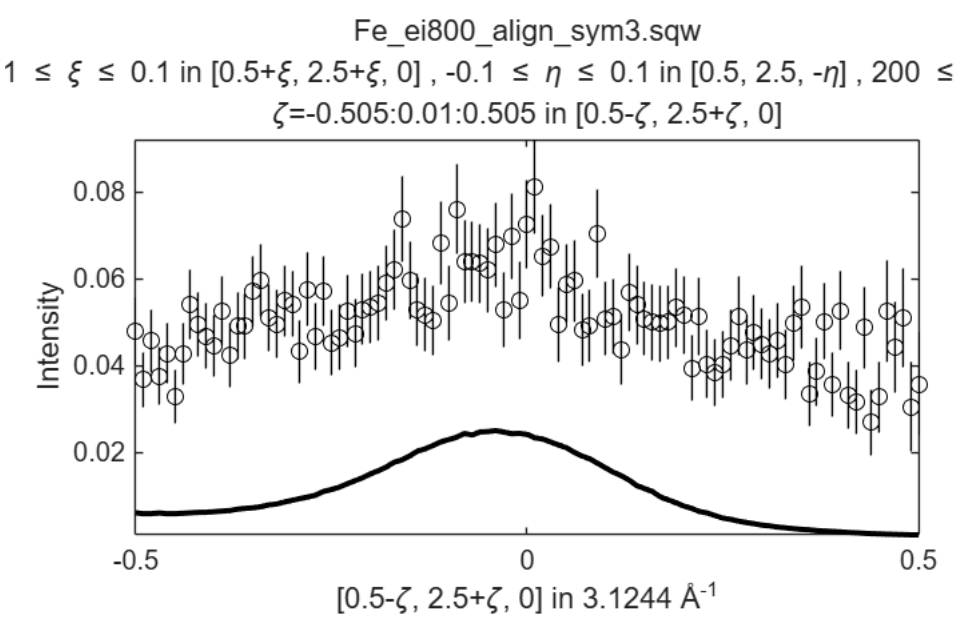
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          238.6   +/-   569.4
    4          2.742   +/-   7.753
    5              0
    6          49.42   +/-   73.63
    7             35   +/-   233.4
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.03827   +/-   0.001941
    2              0
    3              0
```

```
Covariance matrix for free parameters:
-------------------------------------
    1.0000    0.9919    0.9928   -0.9965   -0.4676
    0.9919    1.0000    0.9994   -0.9989   -0.4094
    0.9928    0.9994    1.0000   -0.9993   -0.3926
   -0.9965   -0.9989   -0.9993    1.0000    0.4210
   -0.4676   -0.4094   -0.3926    0.4210    1.0000
```






```
*****************************
**** En = 215±5 going to 405
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.63676s    Reduced Chi^2 = 4.1793
 Free parameter values:
         238.6          2.742          49.42             35        0.04548
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 3.8076s    Reduced Chi^2 = 0.97464      Levenberg-Marquardt = 0.1
```

```
 Free parameter values:
        83.05          1.988          45.98             35        0.03638
--------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 6.7979s    Reduced Chi^2 = 0.89545      Levenberg-Marquardt = 0.01
 Free parameter values:
        160.9          2.128          47.76             35        0.03421
--------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 9.9817s    Reduced Chi^2 = 3.445      Levenberg-Marquardt = 0.001
 Free parameter values:
       -33.01         -2.592          1.292             35        0.03444
 Total time = 10.5932s    Reduced Chi^2 = 0.90619      Levenberg-Marquardt = 0.01
 Free parameter values:
        89.54         0.7009          31.23             35        0.03401
 Total time = 11.2152s    Reduced Chi^2 = 0.85701      Levenberg-Marquardt = 0.1
 Free parameter values:
        136.3          1.807          41.66             35        0.03392
--------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 14.2546s    Reduced Chi^2 = 1.0167      Levenberg-Marquardt = 0.01
 Free parameter values:
        82.11         0.5611          30.55             35        0.03411
 Total time = 14.8476s    Reduced Chi^2 = 0.85303      Levenberg-Marquardt = 0.1
 Free parameter values:
        117.4          1.402          38.15             35         0.0339
--------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 17.9955s    Reduced Chi^2 = 1.289      Levenberg-Marquardt = 0.01
 Free parameter values:
        66.16         0.3181          26.91             35         0.0342
 Total time = 18.7838s    Reduced Chi^2 = 0.84929      Levenberg-Marquardt = 0.1
 Free parameter values:
        102.3           1.06          34.42             35        0.03394
--------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 21.9233s    Reduced Chi^2 = 1.1428      Levenberg-Marquardt = 0.01
 Free parameter values:
        47.66         0.2806           25.1             35        0.03442
 Total time = 22.5334s    Reduced Chi^2 = 0.83774      Levenberg-Marquardt = 0.1
 Free parameter values:
        83.18         0.8107           31.2             35        0.03405
--------------------------------------------------------------------------------
Iteration = 7
------------------
 Total time = 25.5694s    Reduced Chi^2 = 0.95487      Levenberg-Marquardt = 0.01
 Free parameter values:
        41.23         0.3121          24.88             35         0.0346
 Total time = 26.185s    Reduced Chi^2 = 0.83104      Levenberg-Marquardt = 0.1
 Free parameter values:
        68.33         0.6415          28.87             35        0.03416
--------------------------------------------------------------------------------
Iteration = 8
------------------
 Total time = 29.1807s    Reduced Chi^2 = 0.84782      Levenberg-Marquardt = 0.01
 Free parameter values:
        37.82         0.3477          25.16             35        0.03483
 Total time = 29.797s    Reduced Chi^2 = 0.82605      Levenberg-Marquardt = 0.1
 Free parameter values:
```

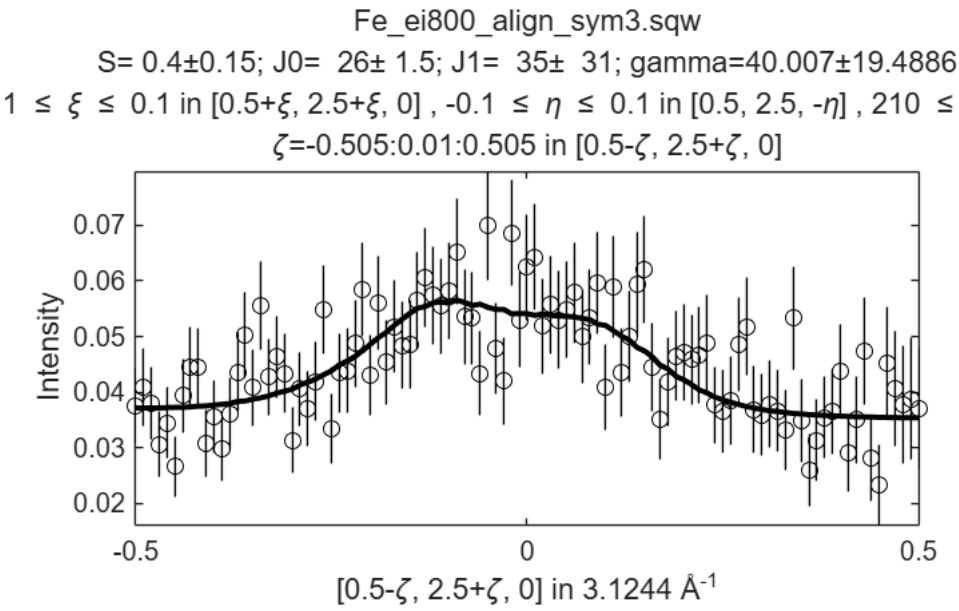
```
         56.35          0.533          27.35             35        0.03435
--------------------------------------------------------------------------------
Iteration = 9
------------------
 Total time = 32.9047s    Reduced Chi^2 = 0.82149      Levenberg-Marquardt = 0.01
 Free parameter values:
         39.33         0.3893          25.72             35        0.03494
--------------------------------------------------------------------------------
Iteration = 10
------------------
 Total time = 35.9586s    Reduced Chi^2 = 0.81438      Levenberg-Marquardt = 0.001
 Free parameter values:
         40.01         0.4042             26             35        0.03504
--------------------------------------------------------------------------------
Iteration = 11
------------------
 Total time = 39.1707s    Reduced Chi^2 = 0.81981      Levenberg-Marquardt = 0.0001
 Free parameter values:
         37.89         0.3897          25.79             35        0.03508
 Total time = 39.7837s    Reduced Chi^2 = 0.82061      Levenberg-Marquardt = 0.001
 Free parameter values:
         37.94         0.3901          25.79             35        0.03508
 Total time = 40.4032s    Reduced Chi^2 = 0.82014      Levenberg-Marquardt = 0.01
 Free parameter values:
          38.3         0.3929          25.82             35        0.03506
 Total time = 41.005s    Reduced Chi^2 = 0.81753      Levenberg-Marquardt = 1
 Free parameter values:
         39.69         0.4028          25.94             35        0.03502
 Total time = 41.6112s    Reduced Chi^2 = 0.81897      Levenberg-Marquardt = 100
 Free parameter values:
         39.97         0.4041          25.99             35        0.03504
 Total time = 42.2424s    Reduced Chi^2 = 0.81835      Levenberg-Marquardt = 10000
 Free parameter values:
            40         0.4042             26             35        0.03504
 *** No improvement in chi-squared over previous iteration ***
 Total time = 42.2508s    Reduced Chi^2 = 0.81438
 Free parameter values:
         40.01         0.4042             26             35        0.03504
--------------------------------------------------------------------------------
Fit converged:

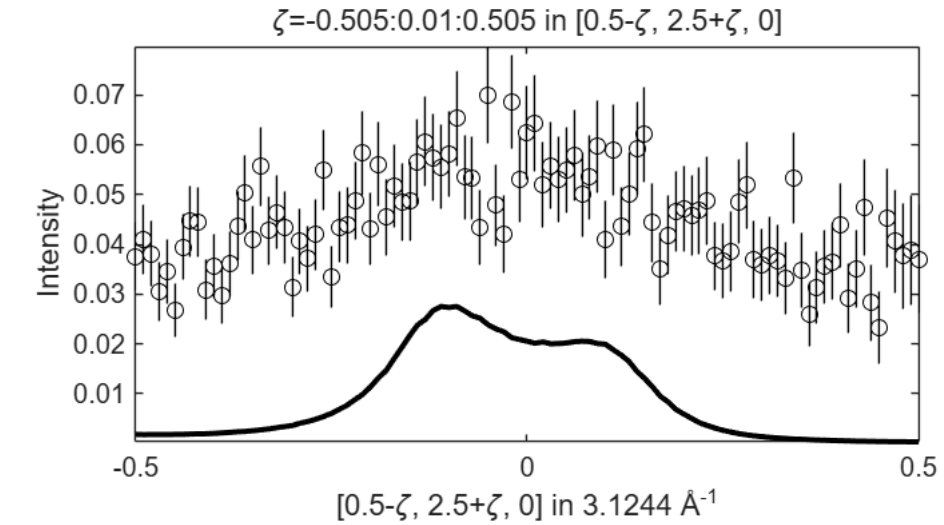
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          40.01   +/-   19.49
    4         0.4042   +/-   0.1536
    5              0
    6             26   +/-   1.525
    7             35   +/-   30.74
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.03504   +/-   0.001573
    2              0
    3              0
```

```
Covariance matrix for free parameters:
-------------------------------------
    1.0000    0.9660    0.8423   -0.9949   -0.6646
    0.9660    1.0000    0.8048   -0.9827   -0.7749
    0.8423    0.8048    1.0000   -0.8631   -0.3721
   -0.9949   -0.9827   -0.8631    1.0000    0.6777
   -0.6646   -0.7749   -0.3721    0.6777    1.0000
```






```
*****************************
**** En = 225±5 going to 405
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.61214s    Reduced Chi^2 = 3.7938
 Free parameter values:
         40.01         0.4042             26             35        0.04194
-------------------------------------------------------------------------------
Iteration = 1
------------------
```

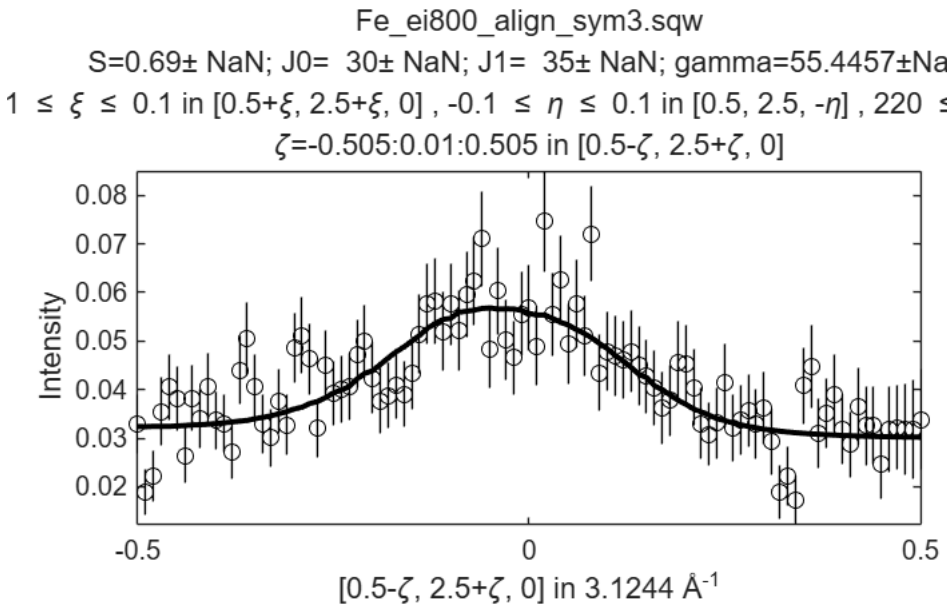
```
 Total time = 3.7428s    Reduced Chi^2 = 1.0174      Levenberg-Marquardt = 0.1
 Free parameter values:
        49.84         0.4825          29.07             35        0.03152
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 6.8841s    Reduced Chi^2 = 0.96218      Levenberg-Marquardt = 0.01
 Free parameter values:
        53.31         0.6384          30.27             35        0.02992
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 10.1228s    Reduced Chi^2 = 0.96162      Levenberg-Marquardt = 0.001
 Free parameter values:
        55.45         0.6858          30.46             35        0.02974
-------------------------------------------------------------------------------
Fit converged:

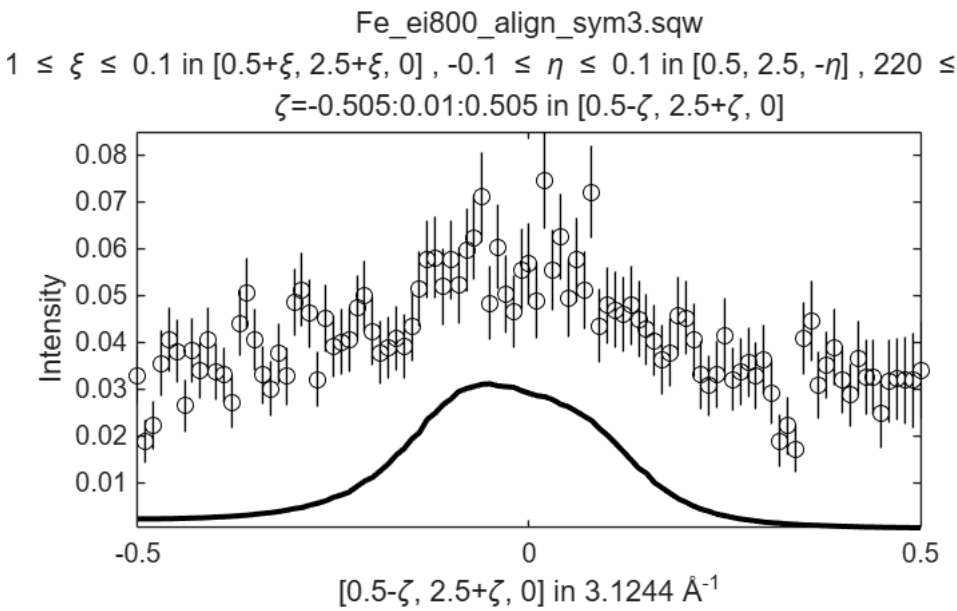
Parameter values (free parameters with error estimates):
-------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          55.45   +/-   NaN
    4         0.6858   +/-   NaN
    5              0
    6          30.46   +/-   NaN
    7             35   +/-   NaN
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.02974   +/-   NaN
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```






```
*****************************
**** En = 235±5 going to 405
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.59205s    Reduced Chi^2 = 4.9567
 Free parameter values:
         55.45         0.6858          30.46             35        0.03762
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 3.7442s    Reduced Chi^2 = 1.0059      Levenberg-Marquardt = 0.1
 Free parameter values:
         64.82         0.7845          33.62             35         0.0267
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 6.957s    Reduced Chi^2 = 1.0089      Levenberg-Marquardt = 0.01
 Free parameter values:
```

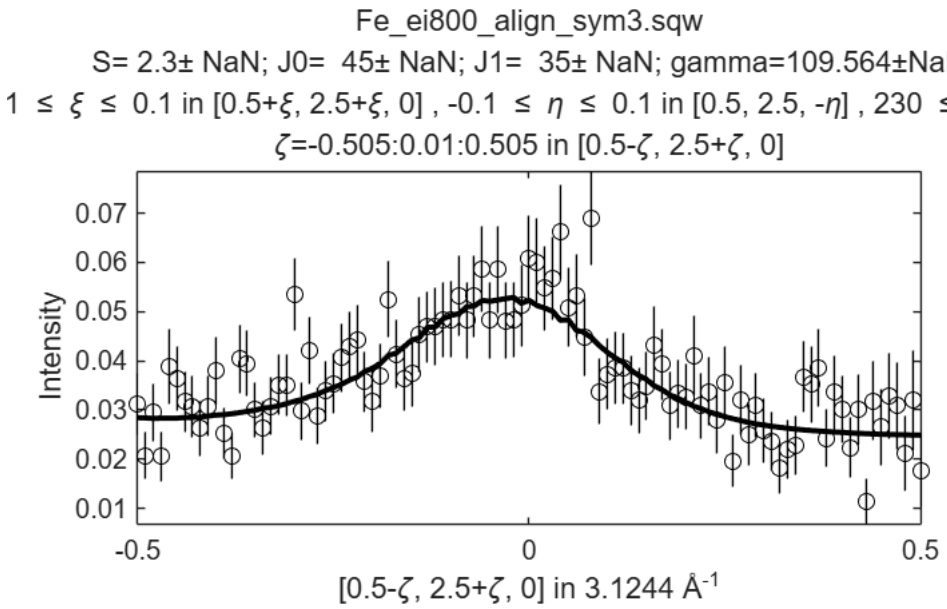
```
        95.52          1.438          39.83             35        0.02461
 Total time = 7.6022s    Reduced Chi^2 = 0.97764      Levenberg-Marquardt = 0.1
 Free parameter values:
        71.54          1.026          35.54             35        0.02536
--------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 10.7336s    Reduced Chi^2 = 0.9802      Levenberg-Marquardt = 0.01
 Free parameter values:
        97.94          1.713          41.33             35        0.02445
 Total time = 11.4378s    Reduced Chi^2 = 0.97179      Levenberg-Marquardt = 0.1
 Free parameter values:
        80.41          1.269          37.29             35        0.02493
--------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 14.6385s    Reduced Chi^2 = 0.97271      Levenberg-Marquardt = 0.01
 Free parameter values:
          104          2.013          43.22             35        0.02431
 Total time = 15.3248s    Reduced Chi^2 = 0.96662      Levenberg-Marquardt = 0.1
 Free parameter values:
        88.79           1.52          39.15             35        0.02466
--------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 18.4764s    Reduced Chi^2 = 0.96598      Levenberg-Marquardt = 0.01
 Free parameter values:
        109.6          2.283          44.78             35        0.02417
--------------------------------------------------------------------------------
Fit converged:

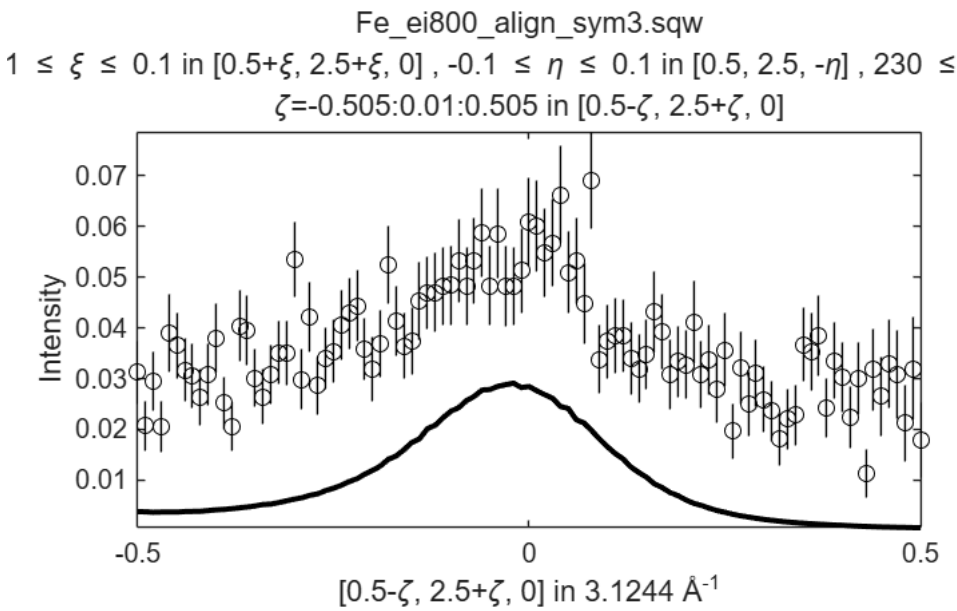
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
-----------------------------
    1              1
    2              8
    3          109.6   +/-   NaN
    4          2.283   +/-   NaN
    5              0
    6          44.78   +/-   NaN
    7             35   +/-   NaN
    8              0
    9              0
   10              0


Background parameter values:
-----------------------------
    1        0.02417   +/-   NaN
    2              0
    3              0


Covariance matrix for free parameters:
---------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```






```
*****************************
**** En = 245±5 going to 405
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.61045s    Reduced Chi^2 = 6.2032
 Free parameter values:
         109.6          2.283          44.78             35         0.0335
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 3.6686s    Reduced Chi^2 = 0.98435      Levenberg-Marquardt = 0.1
 Free parameter values:
         112.4          2.463          48.19             35        0.02193
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 7.0224s    Reduced Chi^2 = 0.95217      Levenberg-Marquardt = 0.01
 Free parameter values:
```

```
        169.8          3.828          56.96             35        0.01963
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 10.45s    Reduced Chi^2 = 1.1031      Levenberg-Marquardt = 0.001
 Free parameter values:
        288.3          9.603          89.36             35        0.01892
 Total time = 11.14s    Reduced Chi^2 = 0.94417      Levenberg-Marquardt = 0.01
 Free parameter values:
        225.5          5.685           67.8             35        0.01929
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 14.4698s    Reduced Chi^2 = 1.0392      Levenberg-Marquardt = 0.001
 Free parameter values:
        358.1          12.68          102.4             35        0.01891
 Total time = 15.1919s    Reduced Chi^2 = 0.94104      Levenberg-Marquardt = 0.01
 Free parameter values:
        279.6           7.91             79             35        0.01914
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 18.4948s    Reduced Chi^2 = 1.0022      Levenberg-Marquardt = 0.001
 Free parameter values:
        414.2          15.91          113.8             35        0.01893
 Total time = 19.1492s    Reduced Chi^2 = 0.93902      Levenberg-Marquardt = 0.01
 Free parameter values:
        329.2          10.46          90.09             35        0.01907
-------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 22.4125s    Reduced Chi^2 = 0.99611      Levenberg-Marquardt = 0.001
 Free parameter values:
        481.6          20.67          129.5             35        0.01891
 Total time = 23.0665s    Reduced Chi^2 = 0.93485      Levenberg-Marquardt = 0.01
 Free parameter values:
        382.3          13.71          102.6             35        0.01902
-------------------------------------------------------------------------------
Iteration = 7
------------------
 Total time = 26.2574s    Reduced Chi^2 = 0.98986      Levenberg-Marquardt = 0.001
 Free parameter values:
        536.2           26.7          145.6             35        0.01891
 Total time = 26.9118s    Reduced Chi^2 = 0.93281      Levenberg-Marquardt = 0.01
 Free parameter values:
        432.5          17.93            116             35        0.01897
-------------------------------------------------------------------------------
Iteration = 8
------------------
 Total time = 30.2591s    Reduced Chi^2 = 0.99905      Levenberg-Marquardt = 0.001
 Free parameter values:
        600.2          35.17          165.5             35        0.01892
 Total time = 30.9805s    Reduced Chi^2 = 0.93702      Levenberg-Marquardt = 0.01
 Free parameter values:
        486.2          23.51          131.6             35        0.01896
 Total time = 31.6761s    Reduced Chi^2 = 0.93601      Levenberg-Marquardt = 0.1
 Free parameter values:
        451.2          19.77          120.8             35        0.01898
 Total time = 32.3397s    Reduced Chi^2 = 0.93577      Levenberg-Marquardt = 10
 Free parameter values:
        435.4          18.15          116.4             35        0.01899
 Total time = 33.0217s    Reduced Chi^2 = 0.93578      Levenberg-Marquardt = 1000
 Free parameter values:
        432.8          17.96          116.1             35        0.01897
```

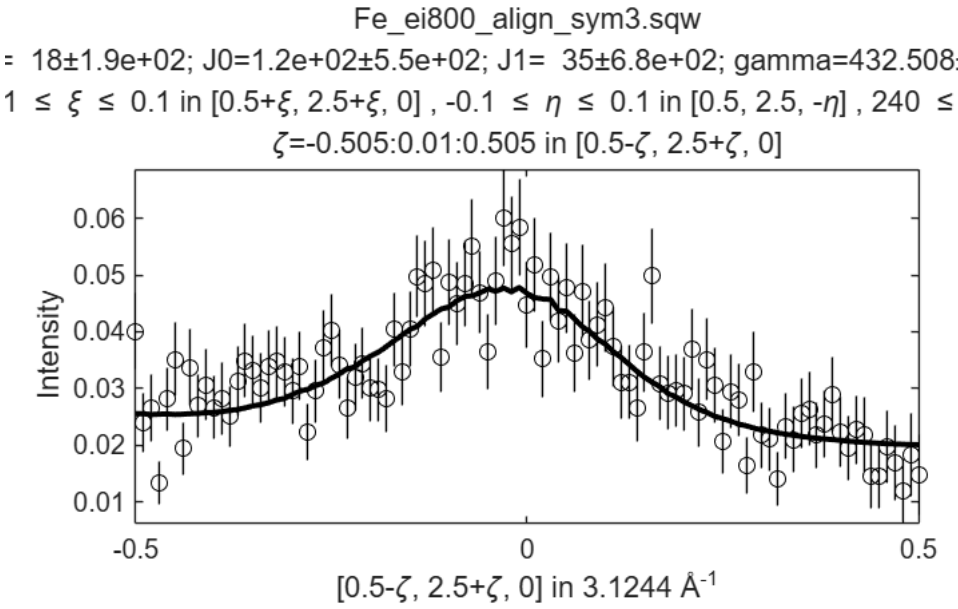
```
 Total time = 33.7212s    Reduced Chi^2 = 0.93383      Levenberg-Marquardt = 100000
 Free parameter values:
        432.5          17.93            116             35        0.01897
 *** No improvement in chi-squared over previous iteration ***
 Total time = 33.7301s    Reduced Chi^2 = 0.93281
 Free parameter values:
        432.5          17.93            116             35        0.01897
--------------------------------------------------------------------------------
Fit converged:

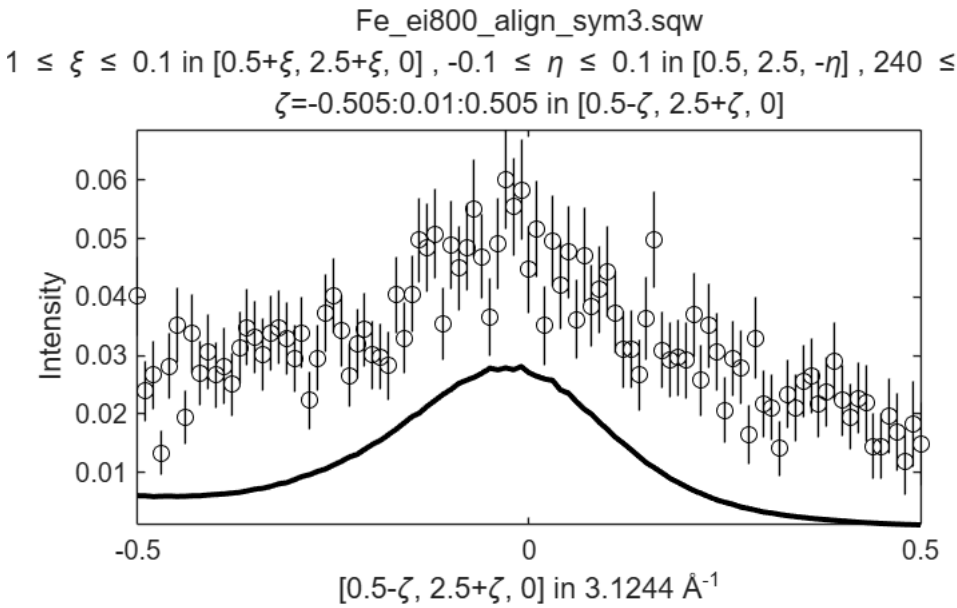
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          432.5   +/-   1981
    4          17.93   +/-   187.3
    5              0
    6            116   +/-   550.6
    7             35   +/-   684
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.01897   +/-   0.001613
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9433    0.9692   -0.9710   -0.6544
    0.9433    1.0000    0.9960   -0.9953   -0.4805
    0.9692    0.9960    1.0000   -1.0000   -0.5300
   -0.9710   -0.9953   -1.0000    1.0000    0.5345
   -0.6544   -0.4805   -0.5300    0.5345    1.0000
```






```
*****************************
**** En = 255±5 going to 405
*****************************
--------------------------------------------------------------------------------
Beginning fit (max 20 iterations)----------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.64639s    Reduced Chi^2 = 8.2306
 Free parameter values:
         432.5          17.93            116             35        0.03008
--------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 3.7993s    Reduced Chi^2 = 1.0707      Levenberg-Marquardt = 0.1
 Free parameter values:
         383.3          15.64          109.8             35        0.01635
--------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 7.2624s    Reduced Chi^2 = 1.0634      Levenberg-Marquardt = 0.01
 Free parameter values:
         362.8          8.806          88.07             35        0.01462
--------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 10.7047s    Reduced Chi^2 = 2.2731      Levenberg-Marquardt = 0.001
 Free parameter values:
         116.4           1.21          44.65             35        0.01475
 Total time = 11.4314s    Reduced Chi^2 = 1.026      Levenberg-Marquardt = 0.01
 Free parameter values:
         271.9          6.831           73.3             35        0.01459
--------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 14.7499s    Reduced Chi^2 = 2.2735      Levenberg-Marquardt = 0.001
 Free parameter values:
         160.2          1.317          47.12             35        0.01489
 Total time = 15.5406s    Reduced Chi^2 = 1.0326      Levenberg-Marquardt = 0.01
 Free parameter values:
         234.3          5.028          64.89             35        0.01476
 Total time = 16.2182s    Reduced Chi^2 = 1.0289      Levenberg-Marquardt = 0.1
 Free parameter values:
```

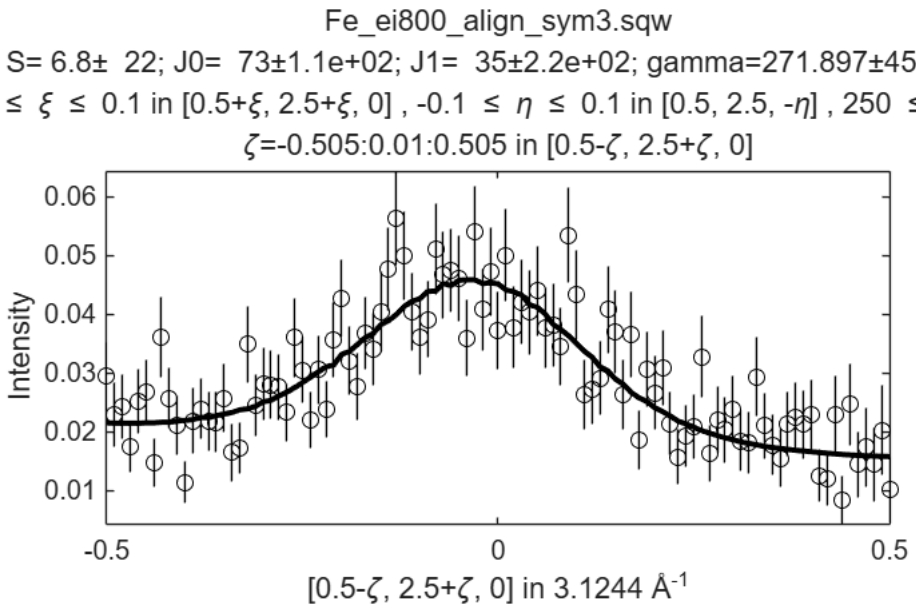
```
        259.1           6.24          70.71             35         0.0147
 Total time = 16.9057s    Reduced Chi^2 = 1.0326      Levenberg-Marquardt = 10
 Free parameter values:
        270.3          6.765          73.07             35         0.0146
 Total time = 17.6715s    Reduced Chi^2 = 1.0301      Levenberg-Marquardt = 1000
 Free parameter values:
        271.7          6.824          73.28             35        0.01459
 Total time = 18.3561s    Reduced Chi^2 = 1.0334      Levenberg-Marquardt = 100000
 Free parameter values:
        271.9           6.83           73.3             35        0.01459
 *** No improvement in chi-squared over previous iteration ***
 Total time = 18.3656s    Reduced Chi^2 = 1.026
 Free parameter values:
        271.9          6.831           73.3             35        0.01459
--------------------------------------------------------------------------------
Fit converged:

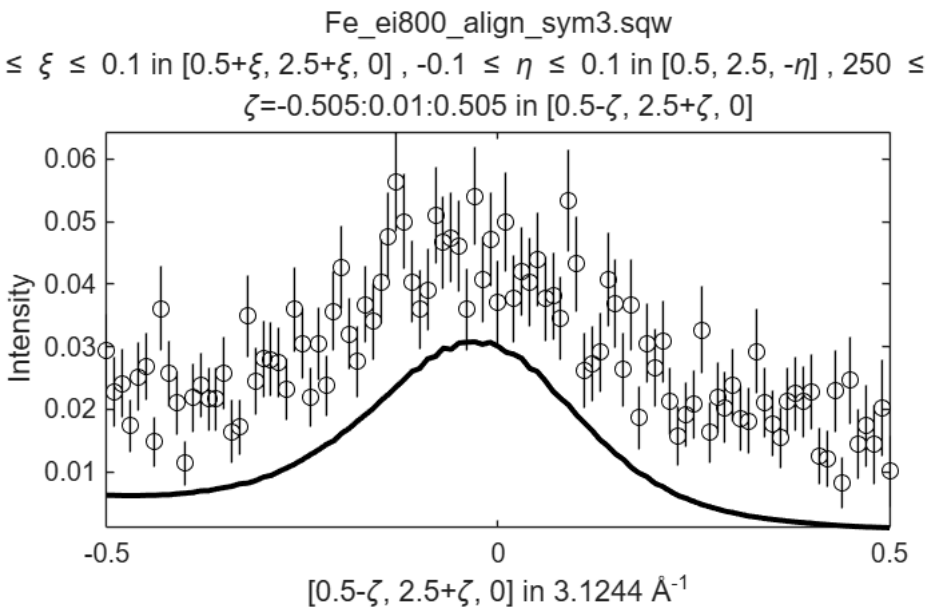
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          271.9   +/-   453.7
    4          6.831   +/-   22.47
    5              0
    6           73.3   +/-   107.1
    7             35   +/-   221.1
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.01459   +/-   0.001538
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9578    0.9734   -0.9776   -0.6204
    0.9578    1.0000    0.9980   -0.9967   -0.4651
    0.9734    0.9980    1.0000   -0.9998   -0.4928
   -0.9776   -0.9967   -0.9998    1.0000    0.5033
   -0.6204   -0.4651   -0.4928    0.5033    1.0000
```






```
*****************************
**** En = 265±5 going to 405
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)---------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.67287s    Reduced Chi^2 = 10.1042
 Free parameter values:
         271.9          6.831           73.3             35        0.02778
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 4.0674s    Reduced Chi^2 = 0.95185      Levenberg-Marquardt = 0.1
 Free parameter values:
         252.9          6.277           74.54            35        0.01452
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 7.619s    Reduced Chi^2 = 0.93399      Levenberg-Marquardt = 0.01
 Free parameter values:
```

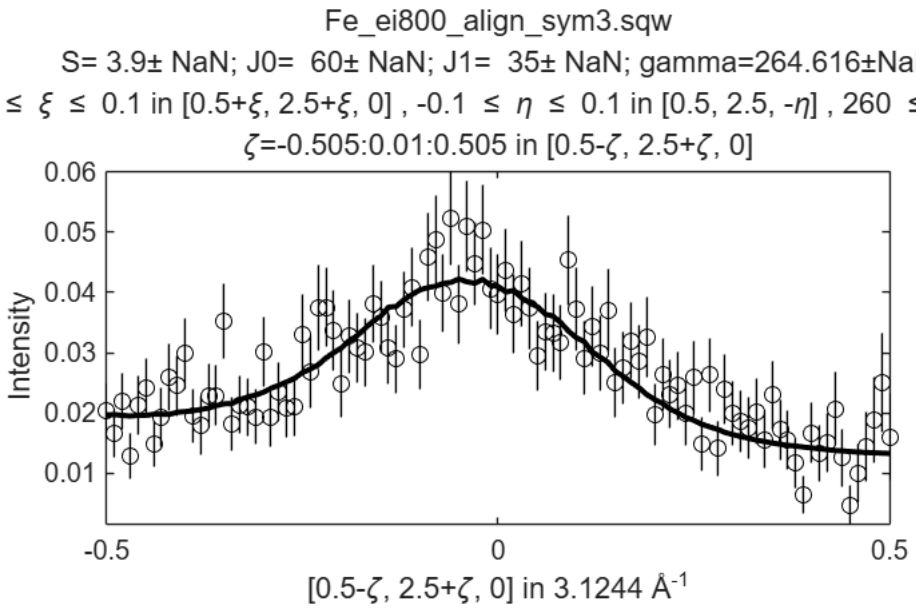
```
        354.9          5.515          75.01             35        0.01228
--------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 11.0784s    Reduced Chi^2 = 0.90467      Levenberg-Marquardt = 0.001
 Free parameter values:
        325.4          5.095          69.18             35        0.01204
--------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 14.5947s    Reduced Chi^2 = 0.90877      Levenberg-Marquardt = 0.0001
 Free parameter values:
        235.5          3.302          56.56             35        0.01212
 Total time = 15.3471s    Reduced Chi^2 = 0.90064      Levenberg-Marquardt = 0.001
 Free parameter values:
        285.7          4.301          63.68             35        0.01207
--------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 19.0434s    Reduced Chi^2 = 0.90204      Levenberg-Marquardt = 0.0001
 Free parameter values:
        240.5          3.388          56.73             35        0.01211
 Total time = 19.7458s    Reduced Chi^2 = 0.9001      Levenberg-Marquardt = 0.001
 Free parameter values:
        264.6          3.866          60.28             35        0.01208
--------------------------------------------------------------------------------
Fit converged:

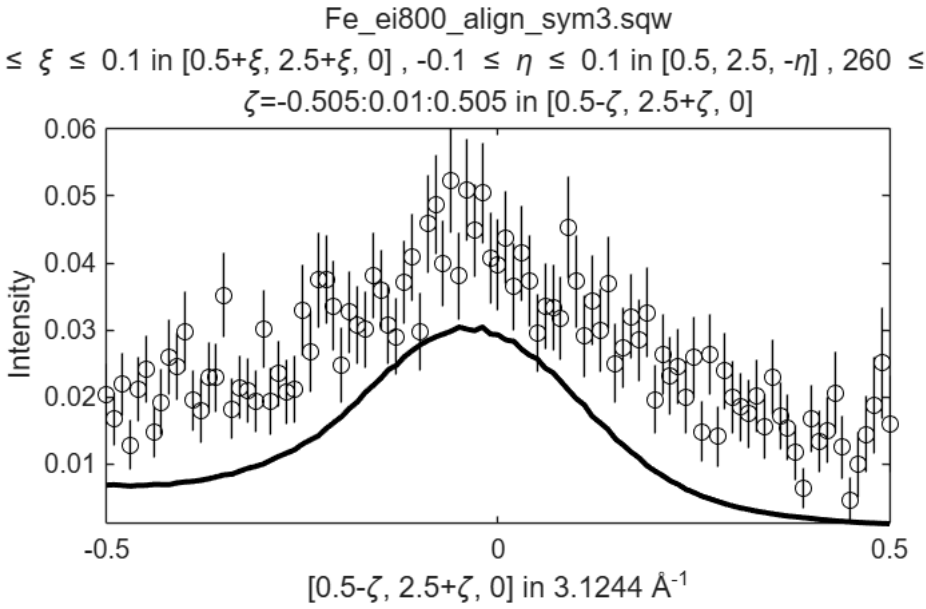
Parameter values (free parameters with error estimates):
-----------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          264.6   +/-   NaN
    4          3.866   +/-   NaN
    5              0
    6          60.28   +/-   NaN
    7             35   +/-   NaN
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.01208   +/-   NaN
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```






```
*****************************
**** En = 275±5 going to 405
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.65729s    Reduced Chi^2 = 11.073
 Free parameter values:
         264.6          3.866          60.28             35        0.02581
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 4.4104s    Reduced Chi^2 = 1.0953      Levenberg-Marquardt = 0.1
 Free parameter values:
         245.2          3.323          63.08             35        0.01408
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 7.9248s    Reduced Chi^2 = 1.0493      Levenberg-Marquardt = 0.01
 Free parameter values:
```

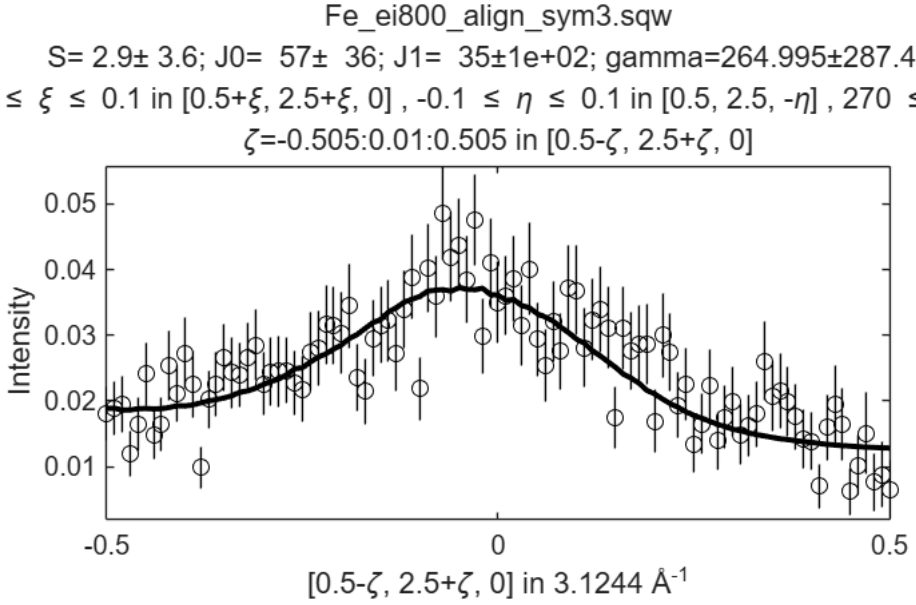
```
         323.3           3.49          64.18             35        0.01175
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 11.4413s    Reduced Chi^2 = 1.0452      Levenberg-Marquardt = 0.001
 Free parameter values:
           286          3.133          59.49             35        0.01164
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 14.9738s    Reduced Chi^2 = 1.0437      Levenberg-Marquardt = 0.0001
 Free parameter values:
         254.1          2.687          55.17             35        0.01166
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 18.4969s    Reduced Chi^2 = 1.0412      Levenberg-Marquardt = 1e-05
 Free parameter values:
         263.7          2.834          56.61             35        0.01167
-------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 22.0329s    Reduced Chi^2 = 1.0443      Levenberg-Marquardt = 1e-06
 Free parameter values:
           272          2.949          57.71             35        0.01165
 Total time = 22.7506s    Reduced Chi^2 = 1.042      Levenberg-Marquardt = 1e-05
 Free parameter values:
           272          2.948           57.7             35        0.01165
 Total time = 23.4732s    Reduced Chi^2 = 1.0437      Levenberg-Marquardt = 0.0001
 Free parameter values:
         271.2          2.938          57.61             35        0.01166
 Total time = 24.1971s    Reduced Chi^2 = 1.0399      Levenberg-Marquardt = 0.01
 Free parameter values:
           265           2.86          56.82             35        0.01167
-------------------------------------------------------------------------------
Iteration = 7
------------------
 Total time = 27.9121s    Reduced Chi^2 = 1.041      Levenberg-Marquardt = 0.001
 Free parameter values:
         266.1          2.873          56.95             35        0.01167
 Total time = 28.6754s    Reduced Chi^2 = 1.0452      Levenberg-Marquardt = 0.01
 Free parameter values:
         265.4          2.865          56.87             35        0.01167
 Total time = 29.4306s    Reduced Chi^2 = 1.0419      Levenberg-Marquardt = 0.1
 Free parameter values:
         265.2          2.862          56.84             35        0.01167
 Total time = 30.1624s    Reduced Chi^2 = 1.0409      Levenberg-Marquardt = 10
 Free parameter values:
           265           2.86          56.82             35        0.01167
 Total time = 30.9058s    Reduced Chi^2 = 1.0405      Levenberg-Marquardt = 1000
 Free parameter values:
           265           2.86          56.82             35        0.01167
 Total time = 31.6315s    Reduced Chi^2 = 1.0439      Levenberg-Marquardt = 100000
 Free parameter values:
           265           2.86          56.82             35        0.01167
 *** No improvement in chi-squared over previous iteration ***
 Total time = 31.6408s    Reduced Chi^2 = 1.0399
 Free parameter values:
           265           2.86          56.82             35        0.01167
-------------------------------------------------------------------------------
Fit converged:

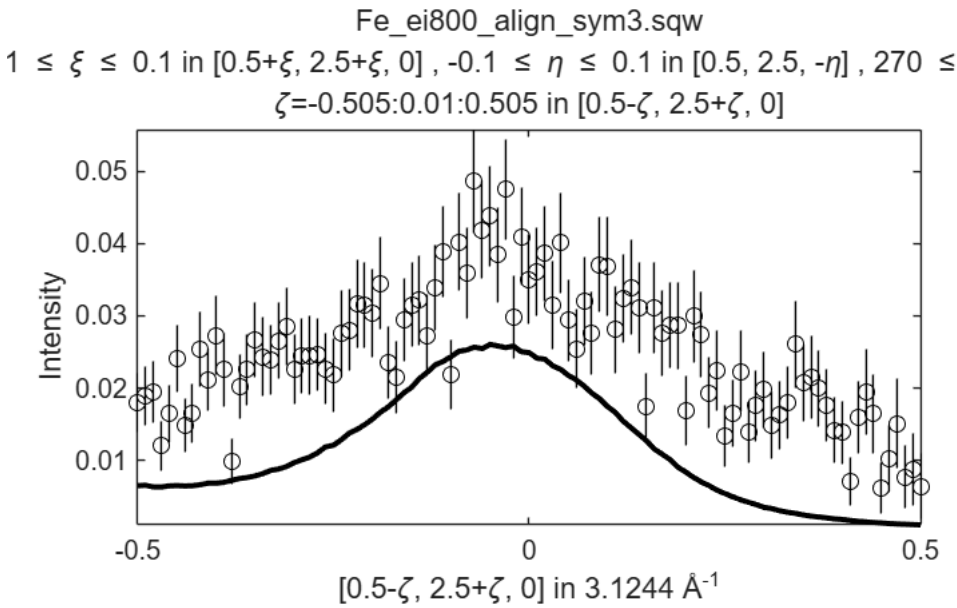
Parameter values (free parameters with error estimates):
---------------------------------------------------------
```

```
Foreground parameter values:
---------------------------
    1              1
    2              8
    3            265   +/-   287.5
    4           2.86   +/-   3.61
    5              0
    6          56.82   +/-   36.08
    7             35   +/-   102.6
    8              0
    9              0
   10              0


Background parameter values:
---------------------------
    1        0.01167   +/-   0.001312
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9906    0.9893   -0.9941   -0.4434
    0.9906    1.0000    0.9982   -0.9983   -0.3988
    0.9893    0.9982    1.0000   -0.9991   -0.3544
   -0.9941   -0.9983   -0.9991    1.0000    0.3743
   -0.4434   -0.3988   -0.3544    0.3743    1.0000
```






```
*****************************
**** En = 285±5 going to 405
*****************************
--------------------------------------------------------------------------------
Beginning fit (max 20 iterations)----------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.72089s    Reduced Chi^2 = 11.1149
 Free parameter values:
           265           2.86          56.82             35        0.02377
--------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 4.5448s    Reduced Chi^2 = 1.3014      Levenberg-Marquardt = 0.1
 Free parameter values:
         282.1          2.365           56.4             35        0.01126
--------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 8.352s    Reduced Chi^2 = 1.261      Levenberg-Marquardt = 0.01
 Free parameter values:
         254.3          1.956          46.16             35       0.008772
--------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 12.2806s    Reduced Chi^2 = 1.334      Levenberg-Marquardt = 0.001
 Free parameter values:
         108.1          1.117          36.34             35       0.008992
 Total time = 13.0751s    Reduced Chi^2 = 1.2356      Levenberg-Marquardt = 0.01
 Free parameter values:
         171.2          1.558          41.67             35       0.008863
--------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 16.925s    Reduced Chi^2 = 1.223      Levenberg-Marquardt = 0.001
 Free parameter values:
         162.1          1.383          39.69             35       0.008937
--------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 20.7482s    Reduced Chi^2 = 1.2226      Levenberg-Marquardt = 0.0001
 Free parameter values:
```

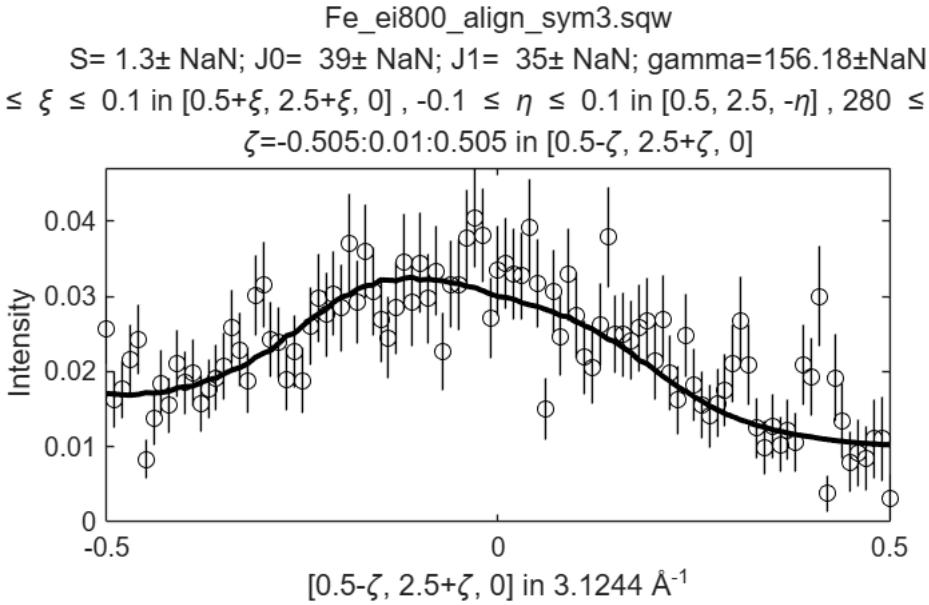
```
         156.2          1.348          39.17             35        0.00895
-------------------------------------------------------------------------------
Fit converged:

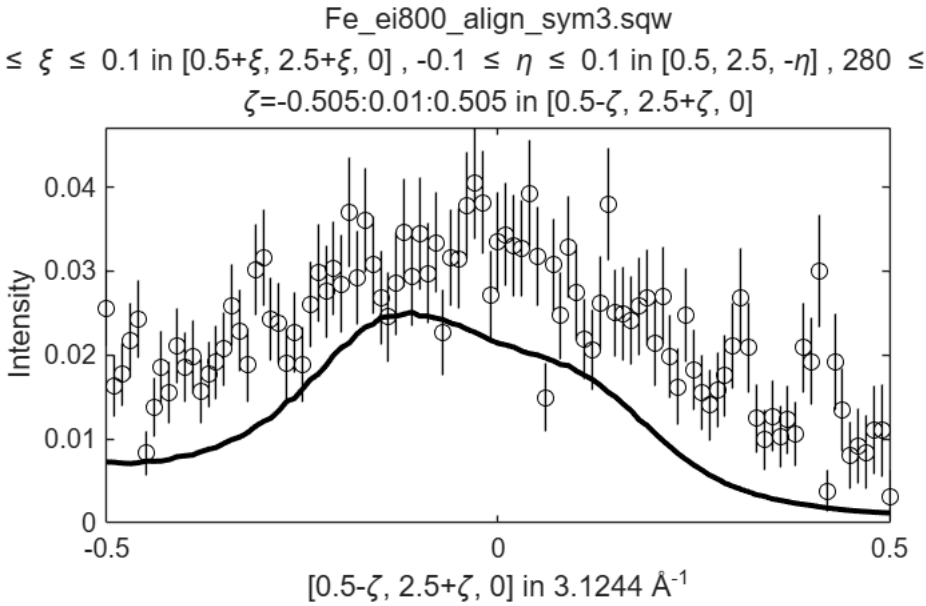
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          156.2   +/-   NaN
    4          1.348   +/-   NaN
    5              0
    6          39.17   +/-   NaN
    7             35   +/-   NaN
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.00895   +/-   NaN
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```






```
*****************************
**** En = 295±5 going to 405
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)---------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.74662s    Reduced Chi^2 = 13.1626
 Free parameter values:
        156.2          1.348          39.17             35        0.02291
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 4.7702s    Reduced Chi^2 = 1.2348      Levenberg-Marquardt = 0.1
 Free parameter values:
        169.4          1.367          42.89             35        0.01031
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 8.9446s    Reduced Chi^2 = 1.1559      Levenberg-Marquardt = 0.01
 Free parameter values:
        228.7          2.007           48.2             35         0.0078
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 13.0605s    Reduced Chi^2 = 1.1437      Levenberg-Marquardt = 0.001
 Free parameter values:
        312.8          2.818           56.5             35        0.00747
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 17.2253s    Reduced Chi^2 = 1.136      Levenberg-Marquardt = 0.0001
 Free parameter values:
        392.7          3.528          63.87             35        0.00743
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 21.3827s    Reduced Chi^2 = 1.139      Levenberg-Marquardt = 1e-05
 Free parameter values:
        403.4          3.593          64.45             35       0.007446
 Total time = 22.1786s    Reduced Chi^2 = 1.137      Levenberg-Marquardt = 0.0001
 Free parameter values:
```

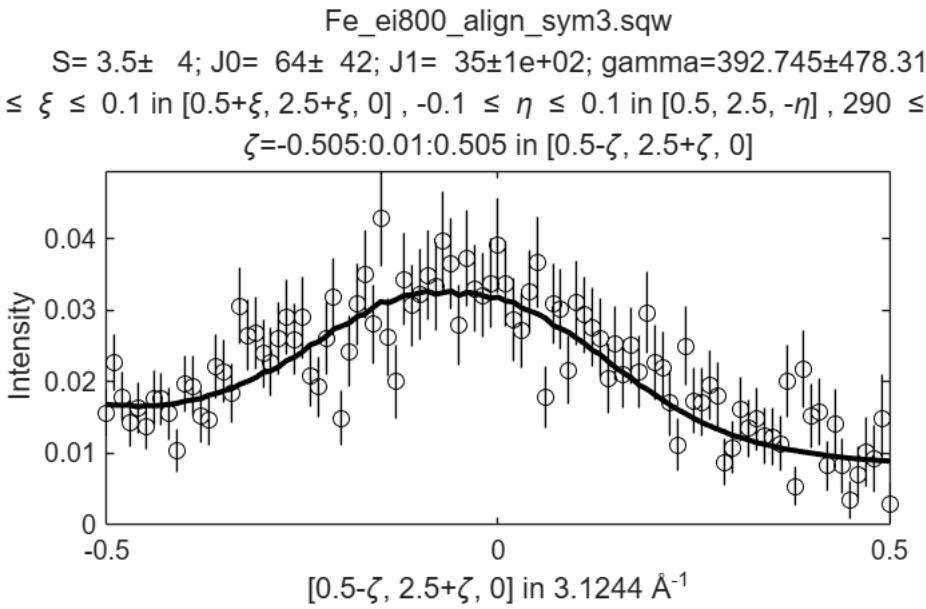
```
        402.7          3.587          64.39             35       0.007447
 Total time = 22.9896s    Reduced Chi^2 = 1.1378      Levenberg-Marquardt = 0.001
 Free parameter values:
        399.9          3.564          64.15             35       0.007449
 Total time = 23.7728s    Reduced Chi^2 = 1.1392      Levenberg-Marquardt = 0.1
 Free parameter values:
        396.3          3.535          63.84             35       0.007452
 Total time = 24.6102s    Reduced Chi^2 = 1.1389      Levenberg-Marquardt = 10
 Free parameter values:
        393.9          3.531          63.84             35       0.007442
 Total time = 25.4571s    Reduced Chi^2 = 1.1368      Levenberg-Marquardt = 1000
 Free parameter values:
        392.9          3.528          63.86             35       0.007431
 *** No improvement in chi-squared over previous iteration ***
 Total time = 25.4641s    Reduced Chi^2 = 1.136
 Free parameter values:
        392.7          3.528          63.87             35        0.00743
--------------------------------------------------------------------------------
Fit converged:

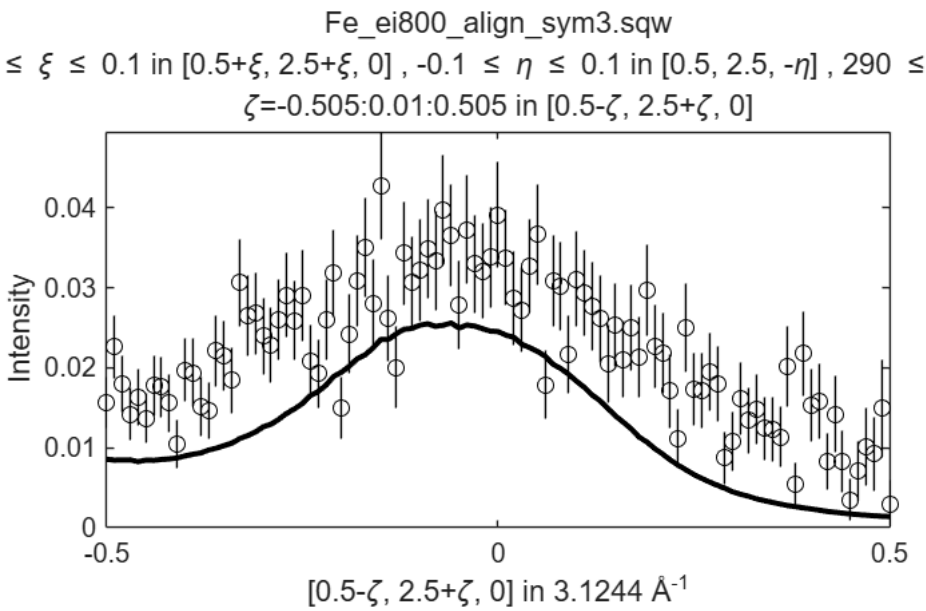
Parameter values (free parameters with error estimates):
-------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          392.7   +/-   478.3
    4          3.528   +/-   3.961
    5              0
    6          63.87   +/-   42.43
    7             35   +/-   103.5
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.00743   +/-   0.001137
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9961    0.9959   -0.9970   -0.3213
    0.9961    1.0000    0.9980   -0.9973   -0.3211
    0.9959    0.9980    1.0000   -0.9996   -0.2700
   -0.9970   -0.9973   -0.9996    1.0000    0.2631
   -0.3213   -0.3211   -0.2700    0.2631    1.0000
```






```
*****************************
**** En = 305±5 going to 405
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)---------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.81911s    Reduced Chi^2 = 14.9541
 Free parameter values:
         392.7          3.528          63.87             35        0.02231
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 5.3064s    Reduced Chi^2 = 1.3167      Levenberg-Marquardt = 0.1
 Free parameter values:
         301.5          2.251           59.2             35        0.01059
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 9.6827s    Reduced Chi^2 = 2.5016      Levenberg-Marquardt = 0.01
 Free parameter values:
```

```
        102.8          0.644          29.74             35       0.008195
 Total time = 10.5502s    Reduced Chi^2 = 1.2462      Levenberg-Marquardt = 0.1
 Free parameter values:
        254.3          1.849          47.87             35       0.008614
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 14.9425s    Reduced Chi^2 = 1.3304      Levenberg-Marquardt = 0.01
 Free parameter values:
        70.58         0.8113          34.15             35       0.008442
 Total time = 15.8033s    Reduced Chi^2 = 1.1842      Levenberg-Marquardt = 0.1
 Free parameter values:
        176.4           1.51          43.34             35       0.008299
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 20.3252s    Reduced Chi^2 = 1.1687      Levenberg-Marquardt = 0.01
 Free parameter values:
        110.7         0.8867           35.6             35       0.008472
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 24.7348s    Reduced Chi^2 = 1.1252      Levenberg-Marquardt = 0.001
 Free parameter values:
        106.9         0.9241          36.07             35       0.008809
-------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 29.035s    Reduced Chi^2 = 1.1232      Levenberg-Marquardt = 0.0001
 Free parameter values:
        107.6          0.929           36.1             35       0.008792
-------------------------------------------------------------------------------
Iteration = 7
------------------
 Total time = 33.415s    Reduced Chi^2 = 1.1265      Levenberg-Marquardt = 1e-05
 Free parameter values:
        107.7         0.9298           36.1             35       0.008785
 Total time = 34.3142s    Reduced Chi^2 = 1.1222      Levenberg-Marquardt = 0.0001
 Free parameter values:
        107.7         0.9298           36.1             35       0.008785
-------------------------------------------------------------------------------
Fit converged:

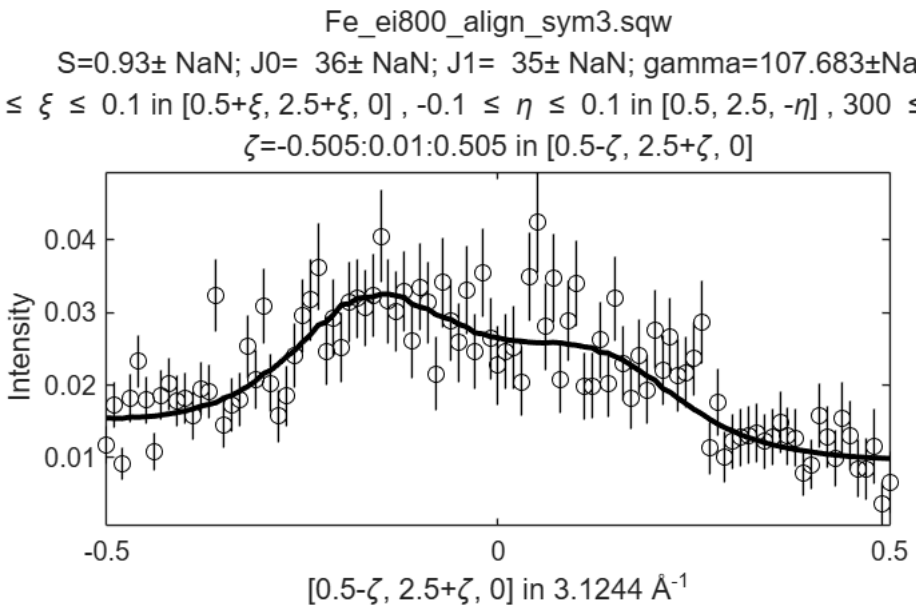
Parameter values (free parameters with error estimates):
---------------------------------------------------------

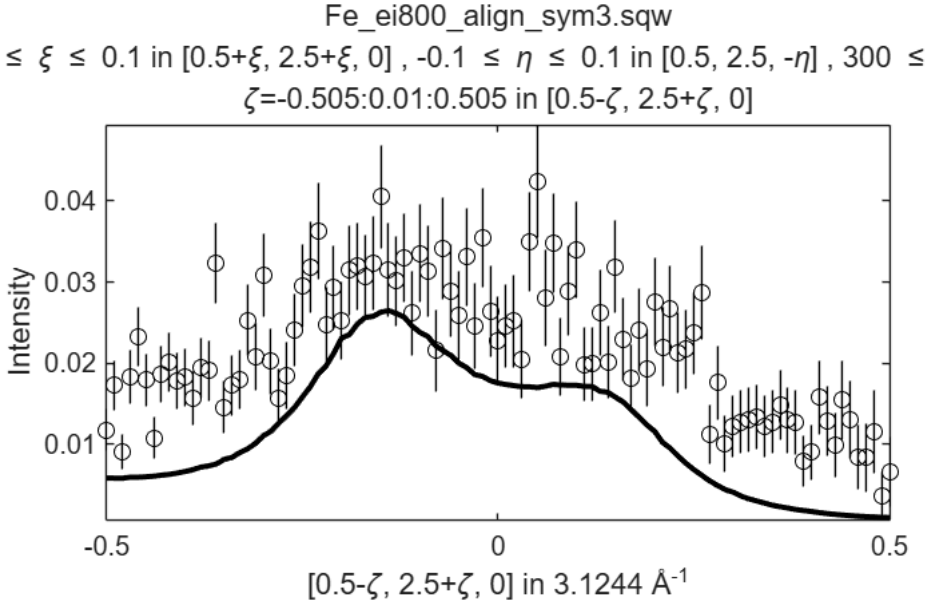
Foreground parameter values:
----------------------------
    1              1
    2              8
    3          107.7   +/-   NaN
    4         0.9298   +/-   NaN
    5              0
    6           36.1   +/-   NaN
    7             35   +/-   NaN
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1       0.008785   +/-   NaN
    2              0
```

```
    3              0


Covariance matrix for free parameters:
---------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```






```
*****************************
**** En = 315±5 going to 405
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.80078s    Reduced Chi^2 = 12.857
 Free parameter values:
         107.7         0.9298           36.1             35         0.0213
-------------------------------------------------------------------------------
```

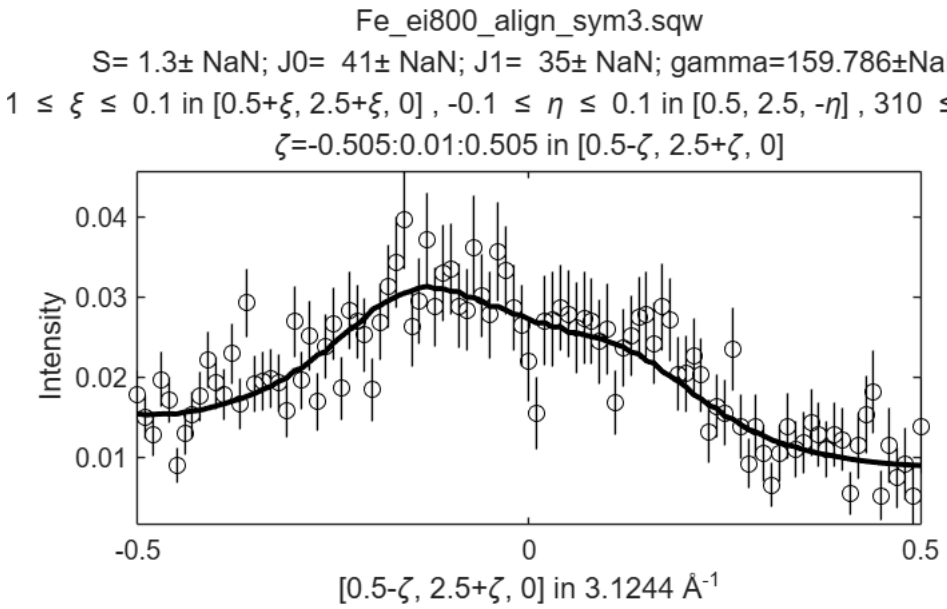
```
Iteration = 1
------------------
 Total time = 5.4086s    Reduced Chi^2 = 1.0839      Levenberg-Marquardt = 0.1
 Free parameter values:
        117.3         0.8757           38.4              35        0.01037
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 10.0584s    Reduced Chi^2 = 1.0084      Levenberg-Marquardt = 0.01
 Free parameter values:
        143.3          1.147          40.13              35       0.008125
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 14.7855s    Reduced Chi^2 = 1.0021      Levenberg-Marquardt = 0.001
 Free parameter values:
        158.1          1.287          41.36              35       0.007814
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 19.4482s    Reduced Chi^2 = 1.006      Levenberg-Marquardt = 0.0001
 Free parameter values:
        159.9          1.306          41.51              35        0.00778
 Total time = 20.3892s    Reduced Chi^2 = 1.0017      Levenberg-Marquardt = 0.001
 Free parameter values:
        159.8          1.305           41.5              35       0.007781
-------------------------------------------------------------------------------
Fit converged:

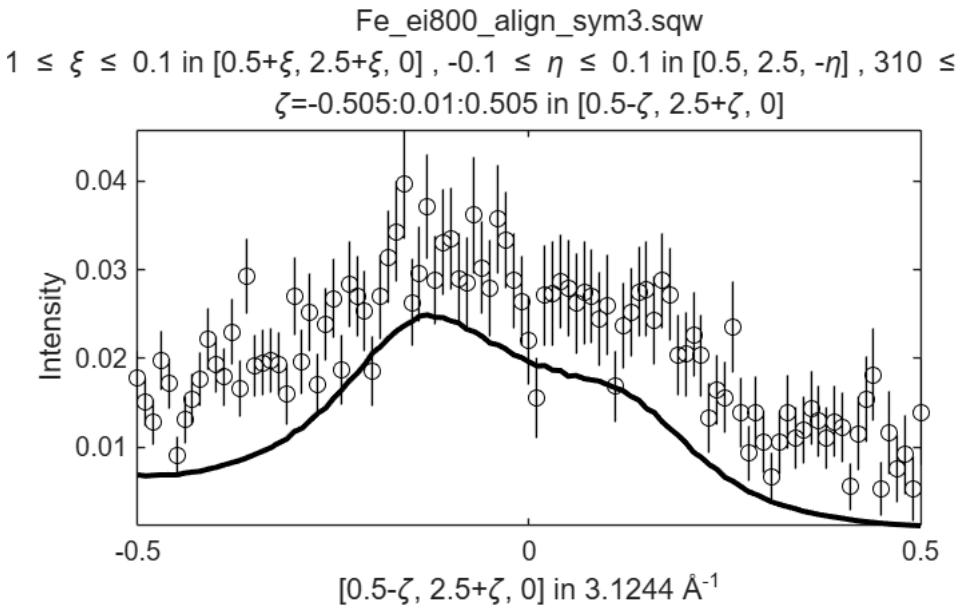
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          159.8   +/-   NaN
    4          1.305   +/-   NaN
    5              0
    6           41.5   +/-   NaN
    7             35   +/-   NaN
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1       0.007781   +/-   NaN
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```






```
*****************************
**** En = 325±5 going to 405
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)---------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 1.054s    Reduced Chi^2 = 16.335
 Free parameter values:
         159.8          1.305           41.5             35        0.02012
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 6.3453s    Reduced Chi^2 = 1.1891      Levenberg-Marquardt = 0.1
 Free parameter values:
         176.8          1.182           42.3             35       0.008373
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 11.6409s    Reduced Chi^2 = 1.0997      Levenberg-Marquardt = 0.01
 Free parameter values:
```

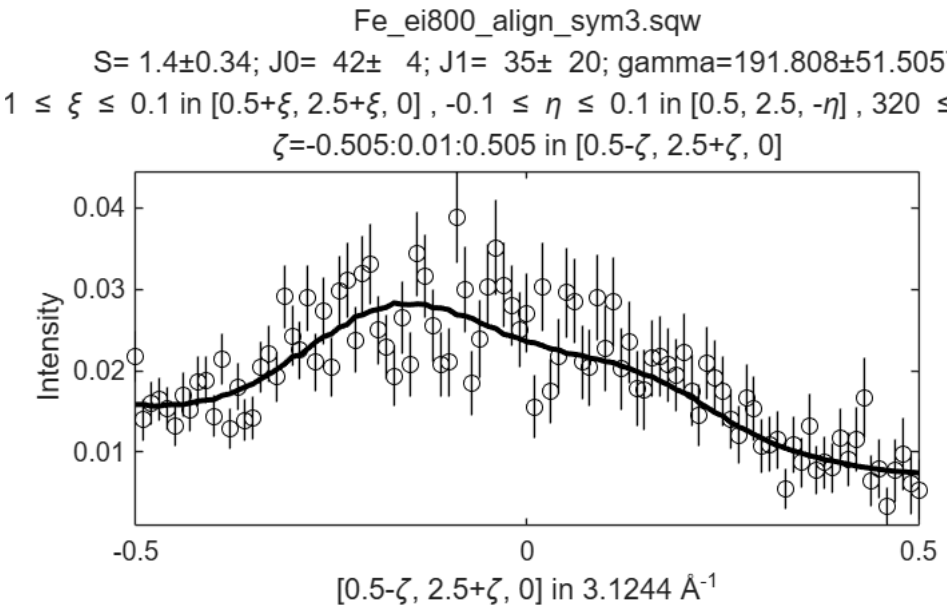
```
        193.8          1.404          42.02             35       0.005976
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 16.9728s    Reduced Chi^2 = 1.0999      Levenberg-Marquardt = 0.001
 Free parameter values:
        191.8          1.415          41.98             35       0.005846
 Total time = 18.0152s    Reduced Chi^2 = 1.1013      Levenberg-Marquardt = 0.01
 Free parameter values:
        191.7          1.414          41.98             35       0.005851
 Total time = 19.1232s    Reduced Chi^2 = 1.0968      Levenberg-Marquardt = 0.1
 Free parameter values:
        191.8          1.412          41.99             35       0.005881
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 24.4891s    Reduced Chi^2 = 1.1037      Levenberg-Marquardt = 0.01
 Free parameter values:
        193.4          1.424          42.09             35       0.005837
 Total time = 25.5671s    Reduced Chi^2 = 1.1024      Levenberg-Marquardt = 0.1
 Free parameter values:
        192.5          1.417          42.02             35       0.005851
 Total time = 26.6441s    Reduced Chi^2 = 1.1009      Levenberg-Marquardt = 1
 Free parameter values:
        192.1          1.414             42             35       0.005868
 Total time = 27.7027s    Reduced Chi^2 = 1.0997      Levenberg-Marquardt = 100
 Free parameter values:
        191.8          1.412             42             35       0.005879
 Total time = 28.8806s    Reduced Chi^2 = 1.1006      Levenberg-Marquardt = 10000
 Free parameter values:
        191.8          1.412          41.99             35       0.005881
 Total time = 29.9488s    Reduced Chi^2 = 1.1014      Levenberg-Marquardt = 1000000
 Free parameter values:
        191.8          1.412          41.99             35       0.005881
 *** No improvement in chi-squared over previous iteration ***
 Total time = 29.959s    Reduced Chi^2 = 1.0968
 Free parameter values:
        191.8          1.412          41.99             35       0.005881
-------------------------------------------------------------------------------
Fit converged:

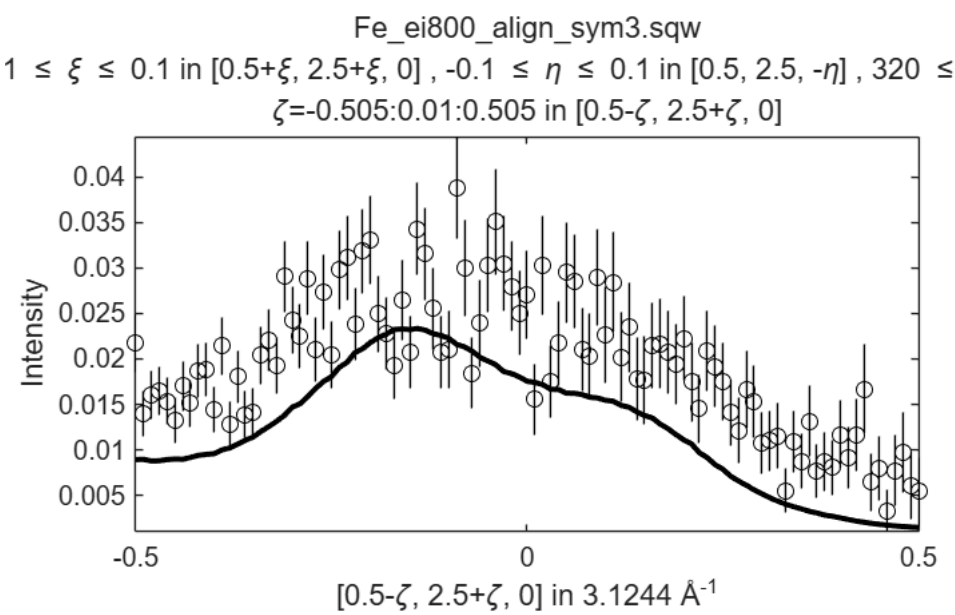
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          191.8   +/-   51.51
    4          1.412   +/-   0.3393
    5              0
    6          41.99   +/-   4.039
    7             35   +/-   19.55
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1       0.005881   +/-   0.000953
    2              0
    3              0
```

```
Covariance matrix for free parameters:
-------------------------------------
    1.0000    0.9654    0.9763   -0.9735   -0.3147
    0.9654    1.0000    0.9440   -0.9143   -0.4936
    0.9763    0.9440    1.0000   -0.9868   -0.2118
   -0.9735   -0.9143   -0.9868    1.0000    0.1100
   -0.3147   -0.4936   -0.2118    0.1100    1.0000
```






```
*****************************
**** En = 335±5 going to 405
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)---------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 1.4819s    Reduced Chi^2 = 19.5951
 Free parameter values:
         191.8          1.412          41.99             35        0.01985
-------------------------------------------------------------------------------
Iteration = 1
------------------
```

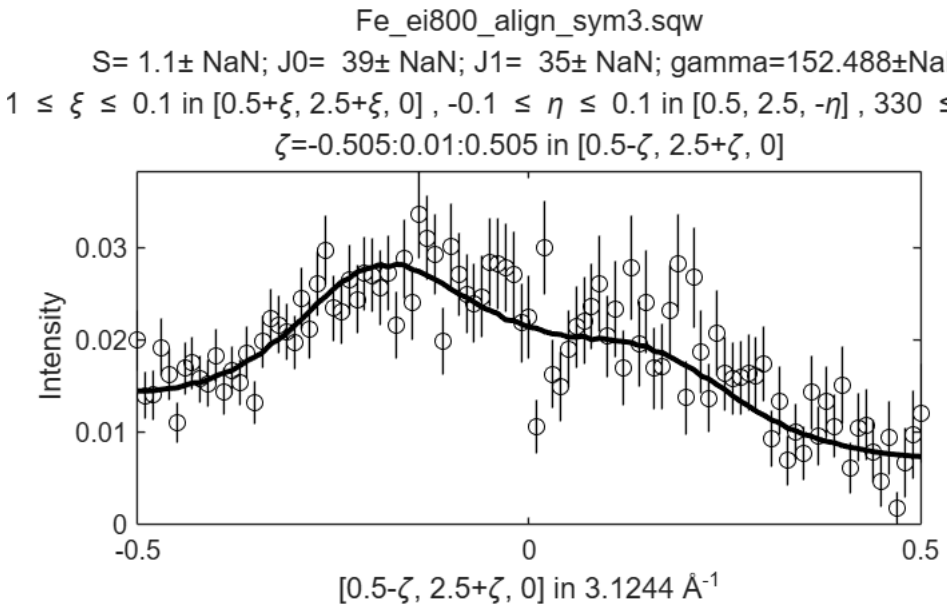
```
 Total time = 9.3137s    Reduced Chi^2 = 1.1862      Levenberg-Marquardt = 0.1
 Free parameter values:
        180.6          1.119          41.74             35       0.007975
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 17.2828s    Reduced Chi^2 = 1.0999      Levenberg-Marquardt = 0.01
 Free parameter values:
        149.9          1.112          38.32             35       0.005895
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 25.1001s    Reduced Chi^2 = 1.0873      Levenberg-Marquardt = 0.001
 Free parameter values:
        152.5          1.115          38.83             35       0.005892
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 32.865s    Reduced Chi^2 = 1.0876      Levenberg-Marquardt = 0.0001
 Free parameter values:
        152.5          1.115          38.85             35       0.005896
 Total time = 34.404s    Reduced Chi^2 = 1.0877      Levenberg-Marquardt = 0.001
 Free parameter values:
        152.5          1.115          38.85             35       0.005896
 Total time = 35.9597s    Reduced Chi^2 = 1.0865      Levenberg-Marquardt = 0.01
 Free parameter values:
        152.5          1.115          38.85             35       0.005896
-------------------------------------------------------------------------------
Fit converged:

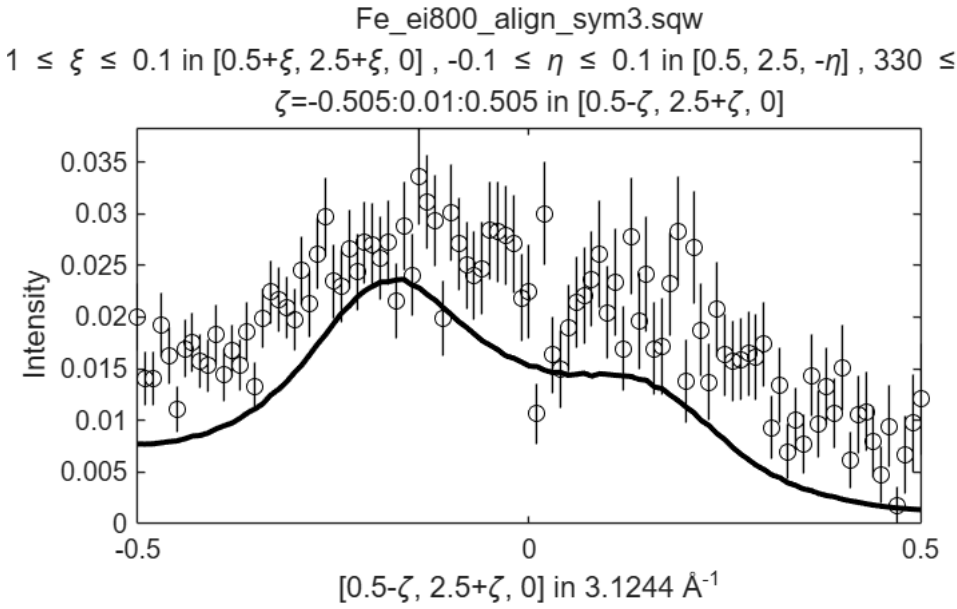
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          152.5   +/-   NaN
    4          1.115   +/-   NaN
    5              0
    6          38.85   +/-   NaN
    7             35   +/-   NaN
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1       0.005896   +/-   NaN
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```






```
*****************************
**** En = 345±5 going to 405
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 1.5879s    Reduced Chi^2 = 19.0537
 Free parameter values:
         152.5          1.115          38.85             35        0.01984
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 9.6652s    Reduced Chi^2 = 1.0371      Levenberg-Marquardt = 0.1
 Free parameter values:
         169.1         0.9149          40.92             35       0.009539
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 17.8366s    Reduced Chi^2 = 0.95925      Levenberg-Marquardt = 0.01
 Free parameter values:
```

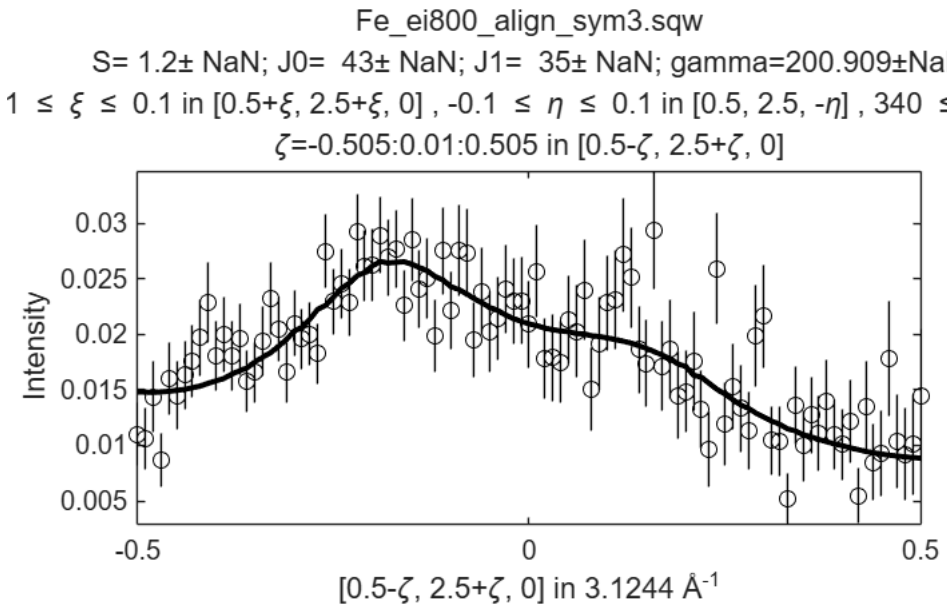
```
        188.4          1.096          41.99             35        0.00763
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 25.936s    Reduced Chi^2 = 0.95652      Levenberg-Marquardt = 0.001
 Free parameter values:
        200.2          1.174          42.74             35       0.007425
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 33.9687s    Reduced Chi^2 = 0.95688      Levenberg-Marquardt = 0.0001
 Free parameter values:
          201          1.179          42.77             35       0.007418
 Total time = 35.5889s    Reduced Chi^2 = 0.95602      Levenberg-Marquardt = 0.001
 Free parameter values:
        200.9          1.179          42.77             35       0.007419
-------------------------------------------------------------------------------
Fit converged:

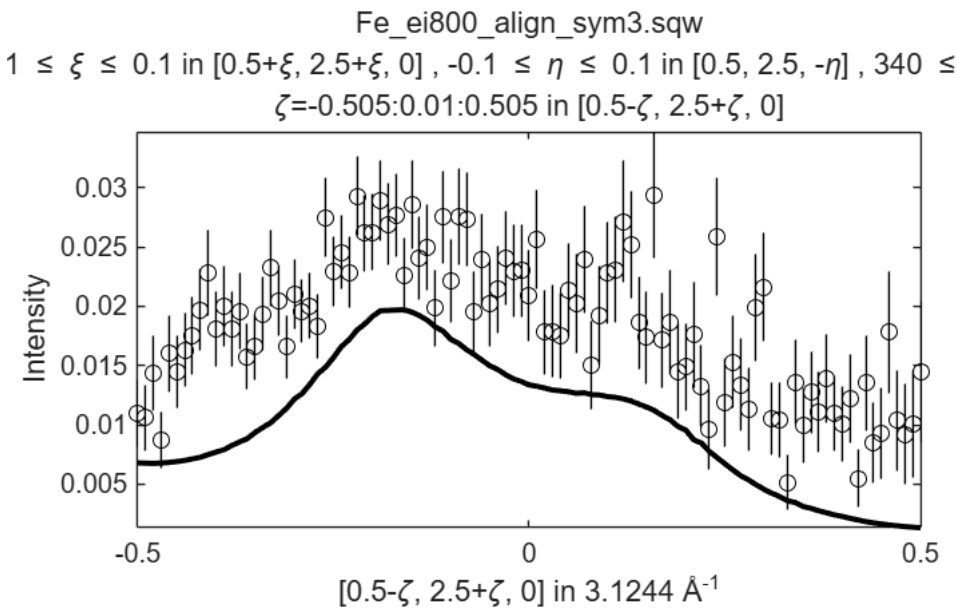
Parameter values (free parameters with error estimates):
-------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          200.9   +/-   NaN
    4          1.179   +/-   NaN
    5              0
    6          42.77   +/-   NaN
    7             35   +/-   NaN
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1       0.007419   +/-   NaN
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```






```
*****************************
**** En = 355±5 going to 405
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 1.5958s    Reduced Chi^2 = 16.5746
 Free parameter values:
         200.9          1.179          42.77             35        0.01962
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 9.6251s    Reduced Chi^2 = 1.1425      Levenberg-Marquardt = 0.1
 Free parameter values:
         179.8          1.006          43.39             35       0.008406
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 17.6625s    Reduced Chi^2 = 1.0468      Levenberg-Marquardt = 0.01
 Free parameter values:
```

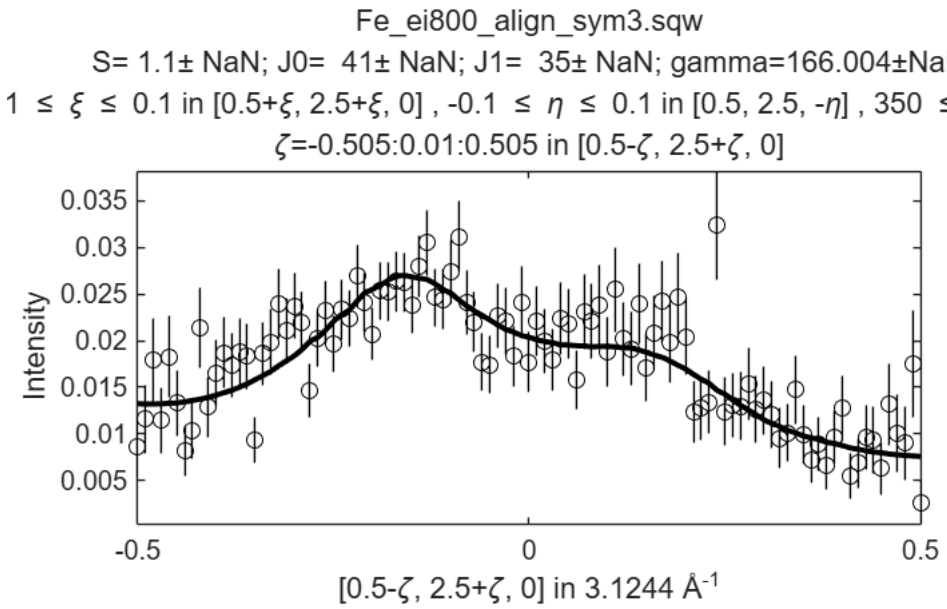
```
         159.8          1.074          40.69             35       0.006293
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 25.7625s    Reduced Chi^2 = 1.0387      Levenberg-Marquardt = 0.001
 Free parameter values:
         166.1          1.104          41.36             35       0.006184
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 33.9097s    Reduced Chi^2 = 1.0385      Levenberg-Marquardt = 0.0001
 Free parameter values:
           166          1.104          41.38             35       0.006186
-------------------------------------------------------------------------------
Fit converged:

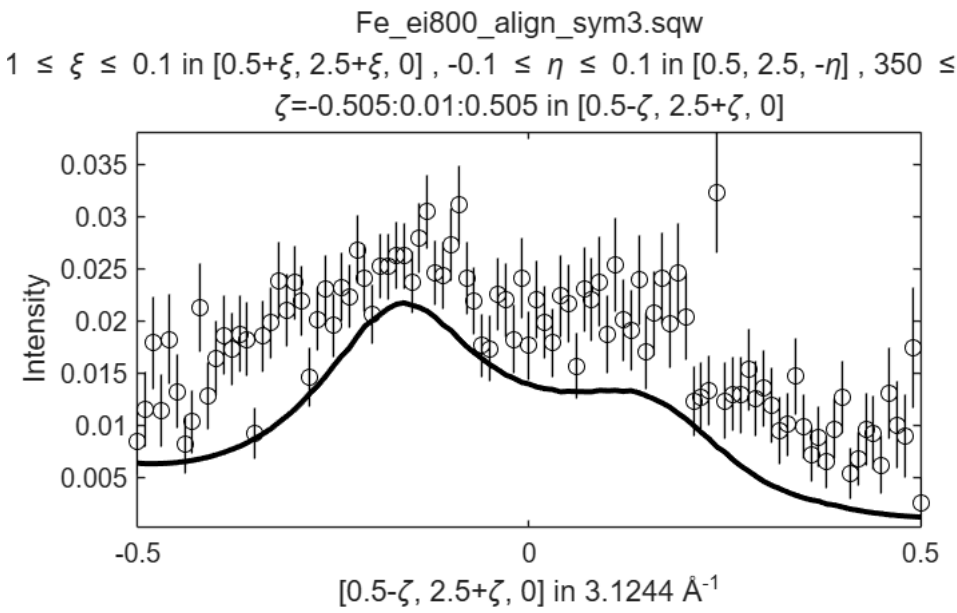
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3            166   +/-   NaN
    4          1.104   +/-   NaN
    5              0
    6          41.38   +/-   NaN
    7             35   +/-   NaN
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1       0.006186   +/-   NaN
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```






```
*****************************
**** En = 365±5 going to 405
*****************************
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.9901
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.99185s    Reduced Chi^2 = 15.4747
 Free parameter values:
           166          1.104          41.38             35        0.01906
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 6.4725s    Reduced Chi^2 = 1.0286      Levenberg-Marquardt = 0.1
 Free parameter values:
         140.8         0.7829          40.53             35       0.008682
-------------------------------------------------------------------------------
Iteration = 2
------------------
```

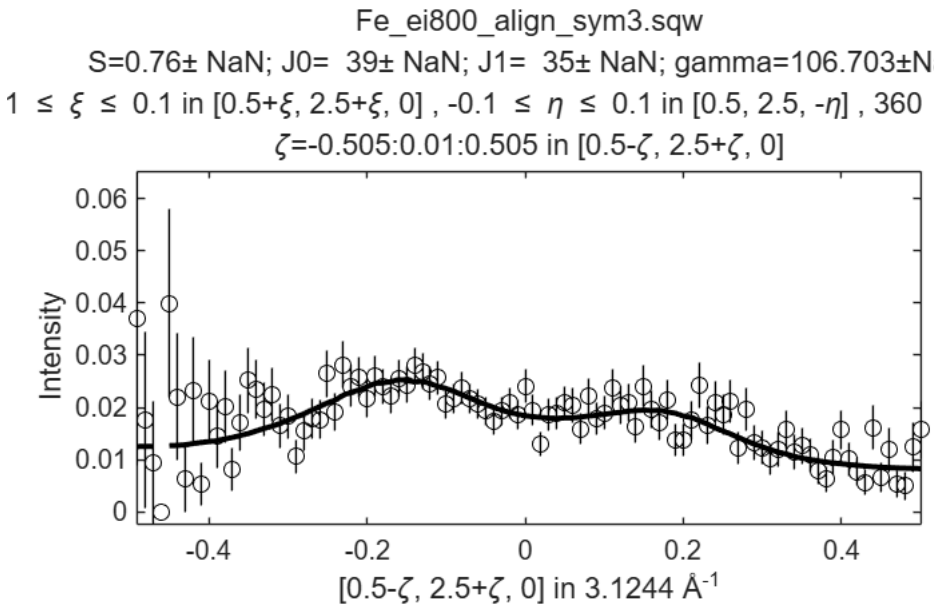
```
 Total time = 11.7963s    Reduced Chi^2 = 0.96618      Levenberg-Marquardt = 0.01
 Free parameter values:
        92.74         0.7264          37.82             35       0.007253
------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 17.204s    Reduced Chi^2 = 0.94051      Levenberg-Marquardt = 0.001
 Free parameter values:
        105.7         0.7593           38.6             35       0.007226
------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 22.5034s    Reduced Chi^2 = 0.93994      Levenberg-Marquardt = 0.0001
 Free parameter values:
        106.7         0.7628          38.74             35       0.007248
------------------------------------------------------------------------------
Fit converged:

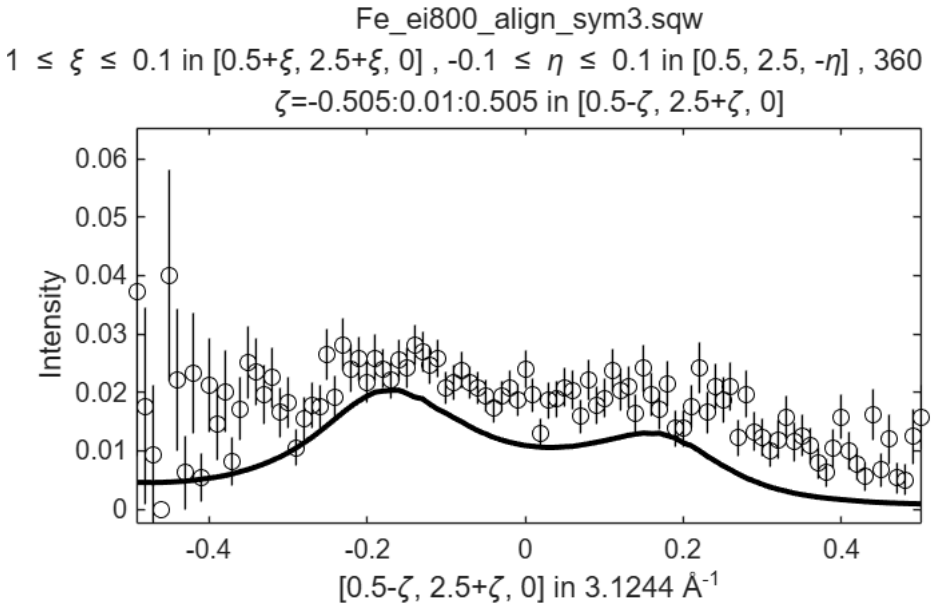
Parameter values (free parameters with error estimates):
-------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          106.7   +/-   NaN
    4         0.7628   +/-   NaN
    5              0
    6          38.74   +/-   NaN
    7             35   +/-   NaN
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1       0.007248   +/-   NaN
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```






```
*****************************
**** En = 375±5 going to 405
*****************************
Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 1.9802
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.95841s    Reduced Chi^2 = 13.1118
 Free parameter values:
         106.7         0.7628          38.74             35         0.0179
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 5.7584s    Reduced Chi^2 = 1.1921      Levenberg-Marquardt = 0.1
 Free parameter values:
         77.01         0.4832          39.89             35       0.009106
-------------------------------------------------------------------------------
Iteration = 2
------------------
```

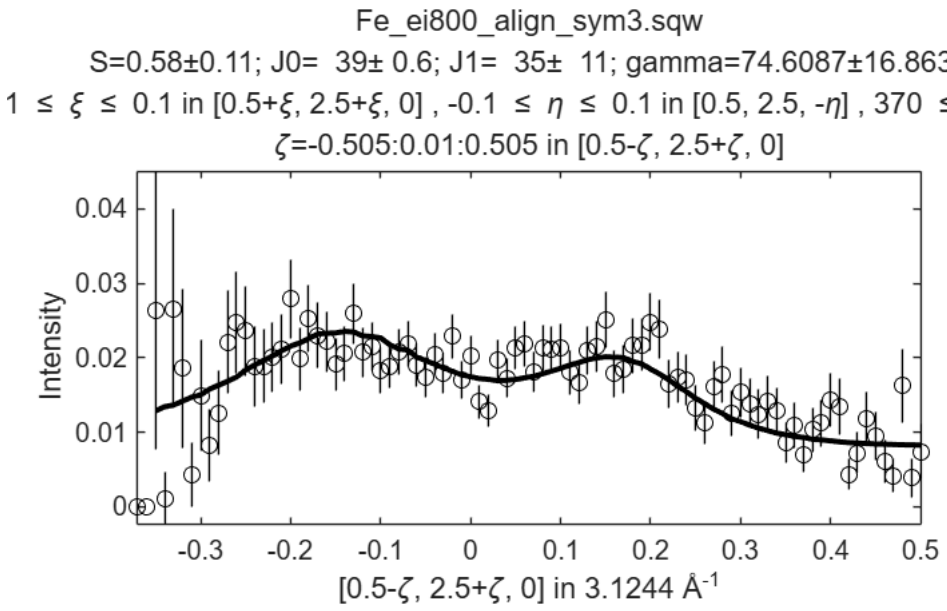
```
 Total time = 10.5997s    Reduced Chi^2 = 1.1238      Levenberg-Marquardt = 0.01
 Free parameter values:
        74.61         0.5785          39.39             35       0.007692
------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 15.5491s    Reduced Chi^2 = 1.1302      Levenberg-Marquardt = 0.001
 Free parameter values:
        76.19         0.5897          39.46             35       0.007587
 Total time = 16.5102s    Reduced Chi^2 = 1.1277      Levenberg-Marquardt = 0.01
 Free parameter values:
        75.99         0.5883          39.46             35       0.007597
 Total time = 17.5027s    Reduced Chi^2 = 1.1266      Levenberg-Marquardt = 0.1
 Free parameter values:
        75.34         0.5838          39.46             35       0.007632
 Total time = 18.4801s    Reduced Chi^2 = 1.1309      Levenberg-Marquardt = 10
 Free parameter values:
        74.67         0.5791          39.41             35       0.007683
 Total time = 19.4438s    Reduced Chi^2 = 1.1247      Levenberg-Marquardt = 1000
 Free parameter values:
        74.61         0.5785           39.4             35       0.007691
 Total time = 20.383s    Reduced Chi^2 = 1.1304      Levenberg-Marquardt = 100000
 Free parameter values:
        74.61         0.5785          39.39             35       0.007692
 *** No improvement in chi-squared over previous iteration ***
 Total time = 20.3912s    Reduced Chi^2 = 1.1238
 Free parameter values:
        74.61         0.5785          39.39             35       0.007692
------------------------------------------------------------------------------
Fit converged:

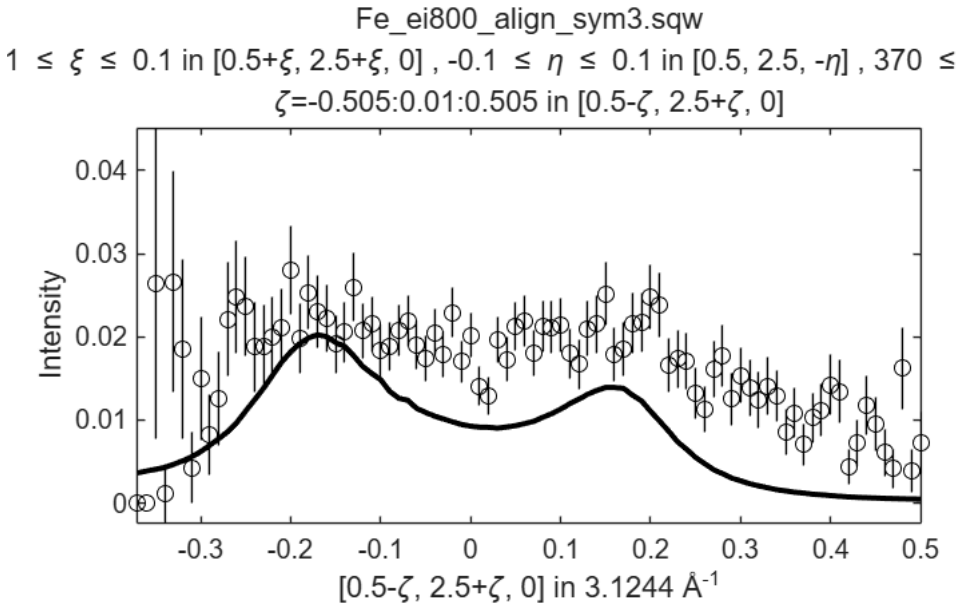
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          74.61   +/-   16.86
    4         0.5785   +/-   0.1068
    5              0
    6          39.39   +/-   0.5954
    7             35   +/-   11.27
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1       0.007692   +/-   0.0009332
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.8884    0.1176   -0.9883   -0.6047
    0.8884    1.0000   -0.0850   -0.8727   -0.8485
    0.1176   -0.0850    1.0000   -0.1224    0.1108
   -0.9883   -0.8727   -0.1224    1.0000    0.5286
   -0.6047   -0.8485    0.1108    0.5286    1.0000
```






```
*****************************
**** En = 385±5 going to 405
*****************************
Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 1.9802
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)---------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.77913s    Reduced Chi^2 = 8.54
 Free parameter values:
         74.61         0.5785          39.39             35          0.016
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 4.9651s    Reduced Chi^2 = 1.0558      Levenberg-Marquardt = 0.1
 Free parameter values:
         76.34         0.4317          39.26             35       0.009926
-------------------------------------------------------------------------------
Iteration = 2
------------------
```

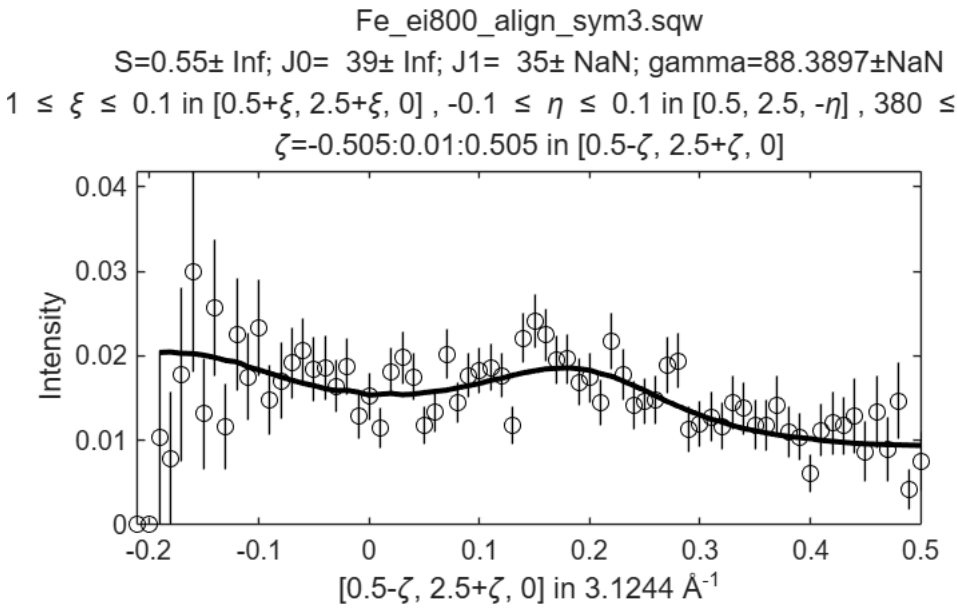
```
 Total time = 9.2547s    Reduced Chi^2 = 1.0377      Levenberg-Marquardt = 0.01
 Free parameter values:
         81.3         0.4975          39.09             35       0.008992
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 13.4734s    Reduced Chi^2 = 1.0365      Levenberg-Marquardt = 0.001
 Free parameter values:
         85.6         0.5294          39.02             35       0.008776
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 17.6644s    Reduced Chi^2 = 1.0363      Levenberg-Marquardt = 0.0001
 Free parameter values:
        88.39         0.5485          38.96             35       0.008658
-------------------------------------------------------------------------------
Fit converged:

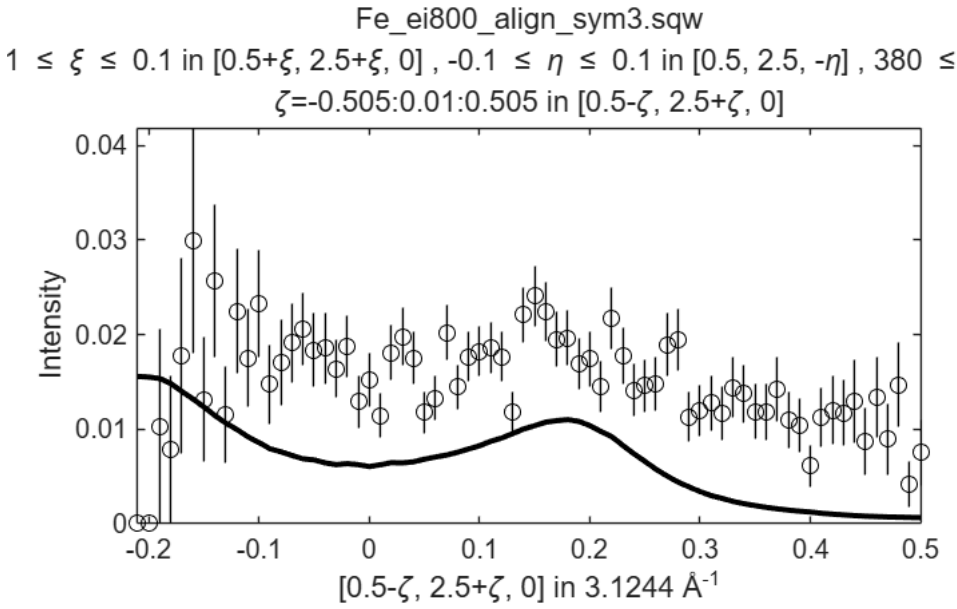
Parameter values (free parameters with error estimates):
-------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          88.39   +/-   NaN
    4         0.5485   +/-   Inf
    5              0
    6          38.96   +/-   Inf
    7             35   +/-   NaN
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1       0.008658   +/-   Inf
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```






```
*****************************
**** En = 395±5 going to 405
*****************************
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.9901
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.72317s    Reduced Chi^2 = 6.6458
 Free parameter values:
         88.39         0.5485          38.96             35        0.01512
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 4.4851s    Reduced Chi^2 = 1.048      Levenberg-Marquardt = 0.1
 Free parameter values:
         73.56         0.4078          40.22             35       0.009587
-------------------------------------------------------------------------------
Iteration = 2
------------------
```

```
 Total time = 8.3514s    Reduced Chi^2 = 1.0191      Levenberg-Marquardt = 0.01
 Free parameter values:
        68.05           0.45          40.01             35       0.008834
------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 12.1974s    Reduced Chi^2 = 1.018      Levenberg-Marquardt = 0.001
 Free parameter values:
        70.06         0.4577          40.07             35       0.008777
------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 16.0592s    Reduced Chi^2 = 1.0154      Levenberg-Marquardt = 0.0001
 Free parameter values:
        70.19         0.4587          40.06             35       0.008769
------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 19.9732s    Reduced Chi^2 = 1.0202      Levenberg-Marquardt = 1e-05
 Free parameter values:
        70.03         0.4591          40.05             35       0.008767
 Total time = 20.8121s    Reduced Chi^2 = 1.0155      Levenberg-Marquardt = 0.0001
 Free parameter values:
        70.03         0.4591          40.05             35       0.008767
 Total time = 21.5733s    Reduced Chi^2 = 1.0165      Levenberg-Marquardt = 0.001
 Free parameter values:
        70.03         0.4591          40.05             35       0.008767
 Total time = 22.3555s    Reduced Chi^2 = 1.0205      Levenberg-Marquardt = 0.1
 Free parameter values:
        70.03         0.4591          40.05             35       0.008768
 Total time = 23.1352s    Reduced Chi^2 = 1.0194      Levenberg-Marquardt = 10
 Free parameter values:
        70.14         0.4588          40.06             35       0.008769
 Total time = 23.9279s    Reduced Chi^2 = 1.0148      Levenberg-Marquardt = 1000
 Free parameter values:
        70.18         0.4587          40.06             35       0.008769
------------------------------------------------------------------------------
Fit converged:

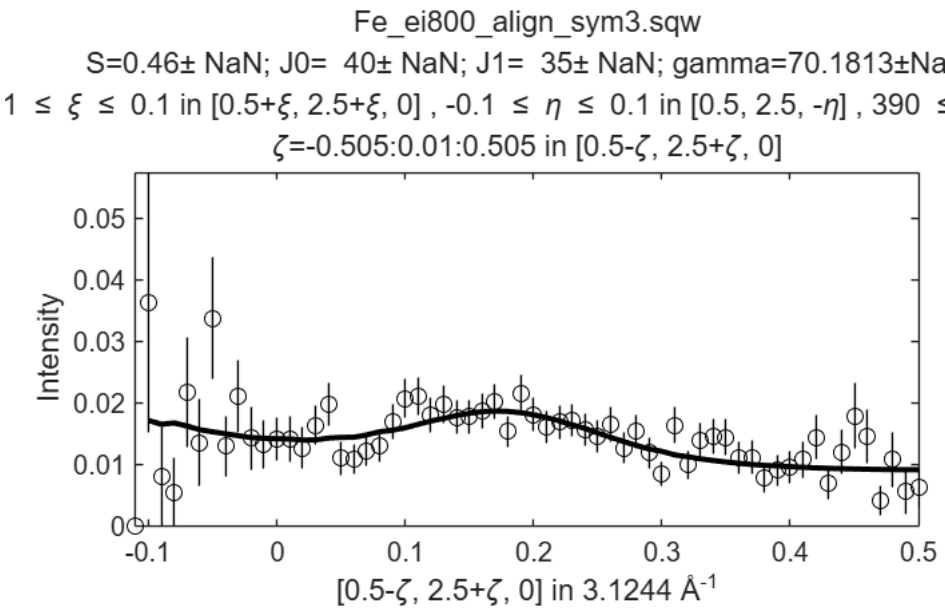
Parameter values (free parameters with error estimates):
---------------------------------------------------------

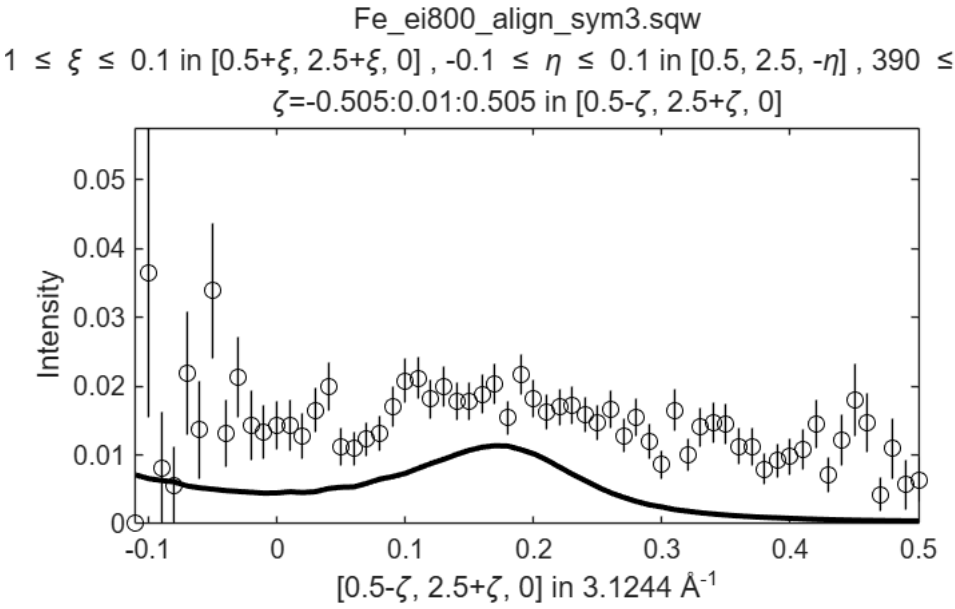
Foreground parameter values:
----------------------------
    1              1
    2              8
    3          70.18   +/-   NaN
    4         0.4587   +/-   NaN
    5              0
    6          40.06   +/-   NaN
    7             35   +/-   NaN
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1       0.008769   +/-   NaN
    2              0
    3              0


Covariance matrix for free parameters:
```

```
------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```






```
*****************************
**** En = 405±5 going to 405
*****************************
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.9901
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.61052s    Reduced Chi^2 = 4.5564
 Free parameter values:
        70.18         0.4587          40.06             35        0.01316
-------------------------------------------------------------------------------
Iteration = 1
------------------
```

```
 Total time = 3.6989s    Reduced Chi^2 = 0.97208      Levenberg-Marquardt = 0.1
 Free parameter values:
        109.5         0.5295          40.79             35       0.008616
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 6.8246s    Reduced Chi^2 = 0.95584      Levenberg-Marquardt = 0.01
 Free parameter values:
        209.4          1.055          42.94             35       0.006243
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 10.0776s    Reduced Chi^2 = 0.99771      Levenberg-Marquardt = 0.001
 Free parameter values:
        395.6          2.282           49.2             35        0.00371
 Total time = 10.7667s    Reduced Chi^2 = 0.89778      Levenberg-Marquardt = 0.01
 Free parameter values:
        297.6          1.716          45.96             35       0.004949
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 13.888s    Reduced Chi^2 = 0.89351      Levenberg-Marquardt = 0.001
 Free parameter values:
          504          3.075          53.45             35       0.003156
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 17.0793s    Reduced Chi^2 = 0.8763      Levenberg-Marquardt = 0.0001
 Free parameter values:
        964.5          5.589          69.74             35       0.001959
-------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 20.3256s    Reduced Chi^2 = 0.95647      Levenberg-Marquardt = 1e-05
 Free parameter values:
         2125          10.04          103.7             35       0.001559
 Total time = 21.0295s    Reduced Chi^2 = 0.9034      Levenberg-Marquardt = 0.0001
 Free parameter values:
         1773          8.583          92.88             35       0.001874
 Total time = 21.6776s    Reduced Chi^2 = 0.86423      Levenberg-Marquardt = 0.001
 Free parameter values:
         1319          6.715          78.93             35       0.002268
-------------------------------------------------------------------------------
Iteration = 7
------------------
 Total time = 25.0056s    Reduced Chi^2 = 0.90427      Levenberg-Marquardt = 0.0001
 Free parameter values:
         2521          10.58          109.7             35       0.001759
 Total time = 25.6478s    Reduced Chi^2 = 0.86359      Levenberg-Marquardt = 0.001
 Free parameter values:
         1772          8.065          89.76             35       0.002101
-------------------------------------------------------------------------------
Fit converged:

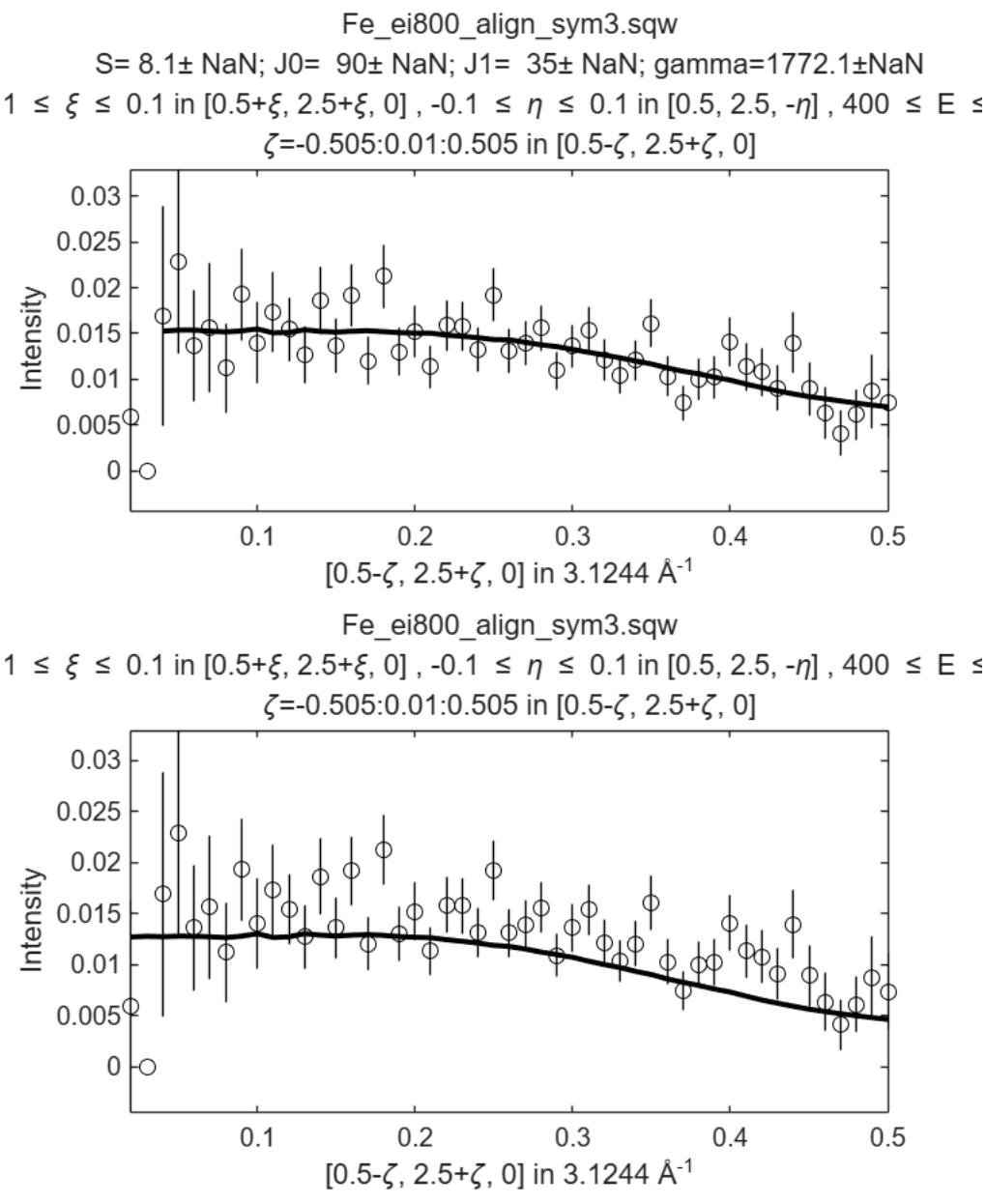
Parameter values (free parameters with error estimates):
-------------------------------------------------------

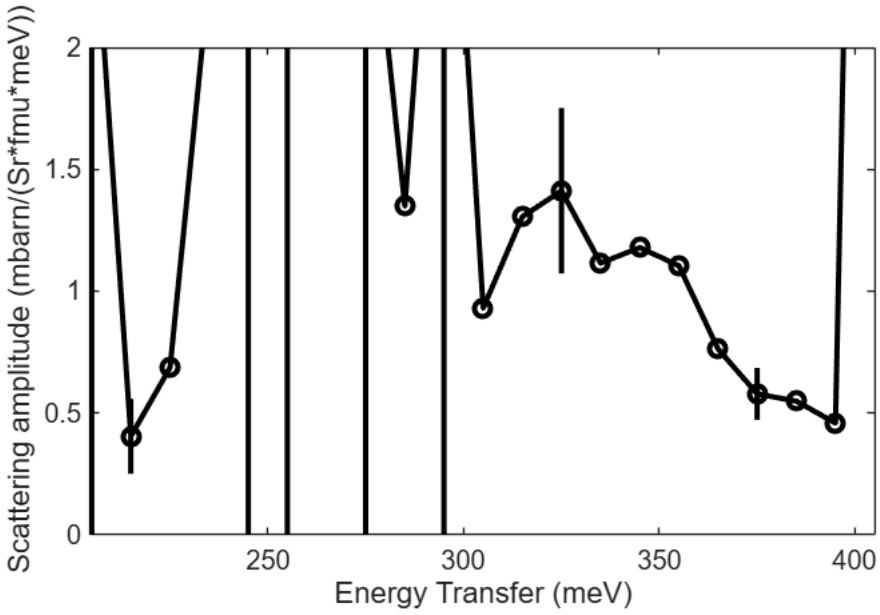
Foreground parameter values:
----------------------------
    1              1
    2              8
    3           1772   +/-   NaN
    4          8.065   +/-   NaN
    5              0
```

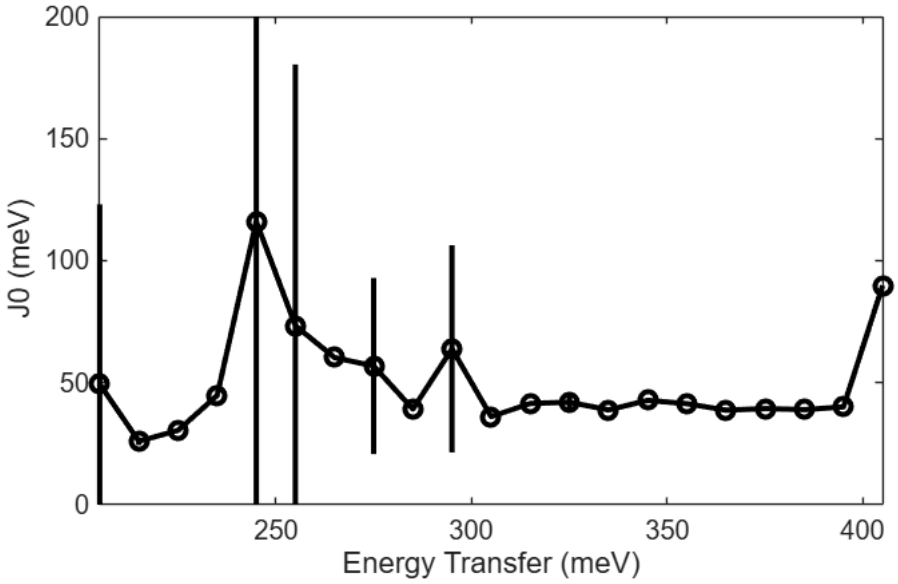
```
    6          89.76   +/-   NaN
    7             35   +/-   NaN
    8              0
    9              0
   10              0

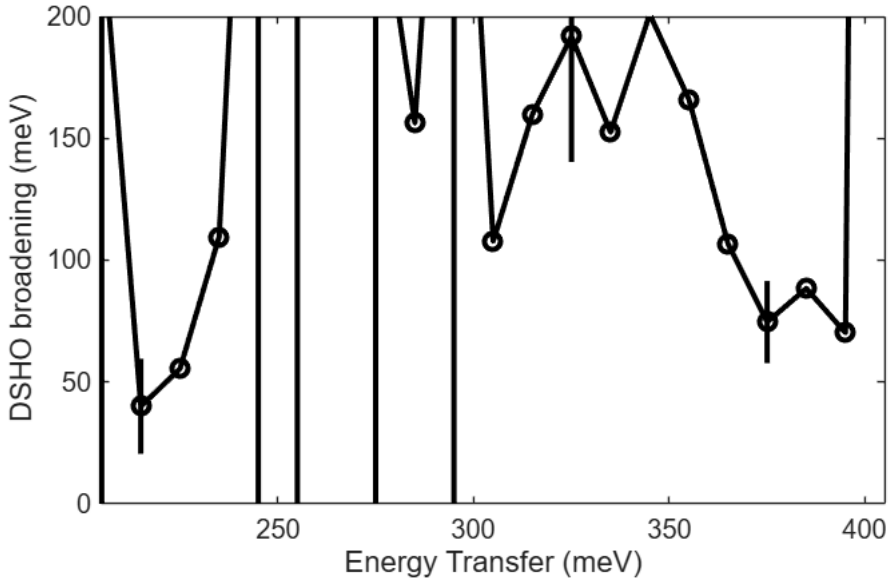

Background parameter values:
----------------------------
    1       0.002101   +/-   NaN
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```




Scattering amplitude (mbarn/(Sr*fmu*meV))
2
1.5
1
0.5
0
250
300
350
400
Energy Transfer (meV)

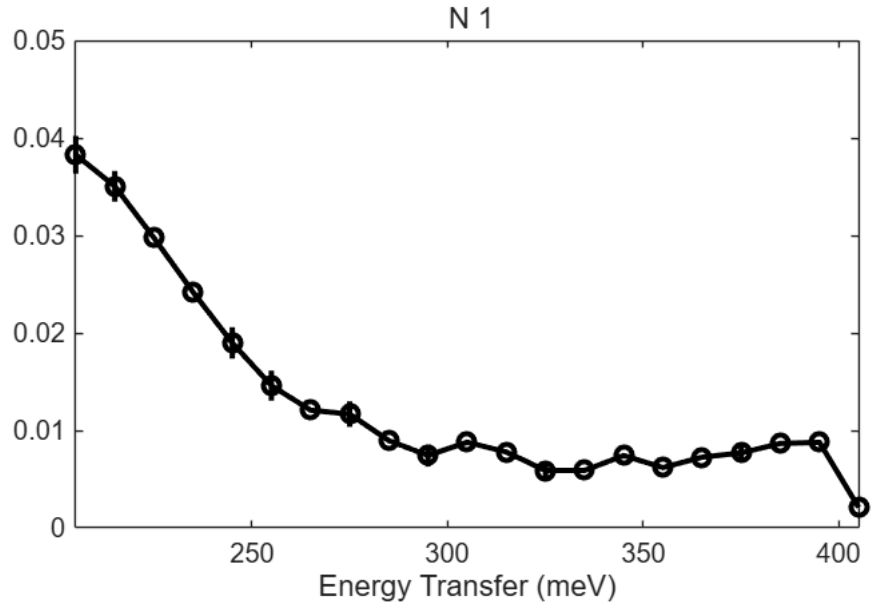

J0 (meV)
200
150
100
50
0
250
300
350
400
Energy Transfer (meV)



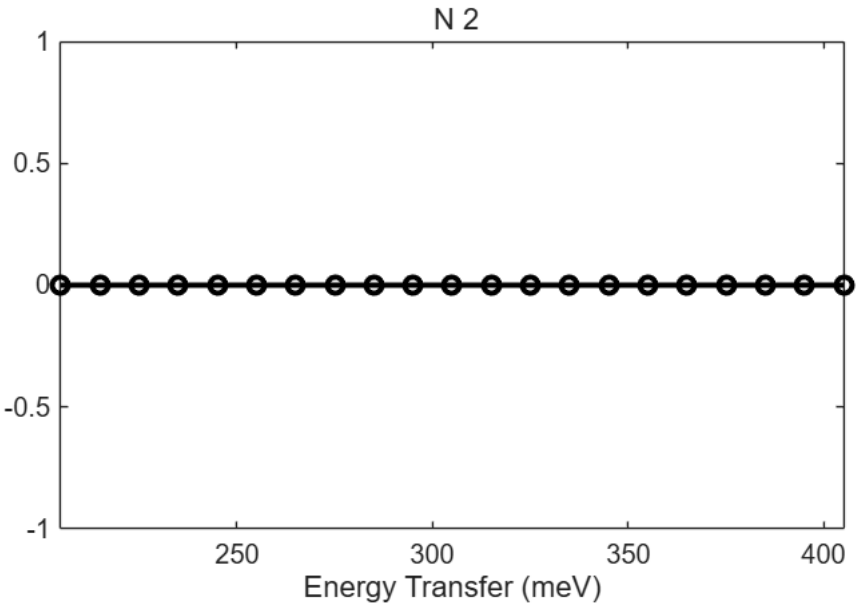

```
cl = HN_FFCutsFits.(selectCuts);
if iscell(cl)
    HN_FFCutsFits.(selectCuts) = [cl{:}];
end

beep
beep
beep
```

```
all_fit_par = HN_FFCutsFits.([selectCuts,'_2Jng_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);ly 0 0.05; keep_figure;
```



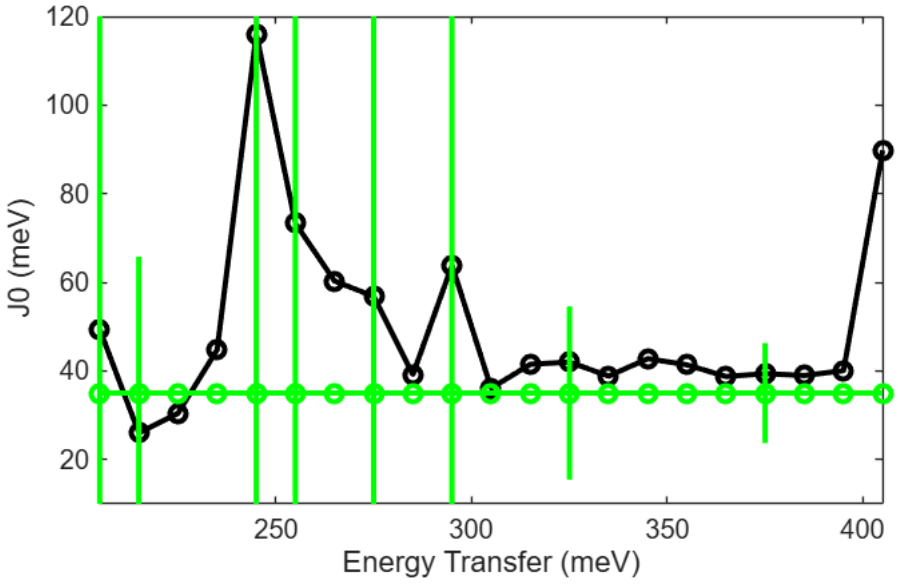

```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```




```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fi
gure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fi
gure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_HN.m';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_HN_',selectCuts,'.pdf']);
export(thisFile,targ_file);
%close all;
%end
```

```
J0 = extract_fit_par(all_fit_par,6,'J0','meV');
J1 = extract_fit_par(all_fit_par,7,'J1','meV');
acolor k; pd(J0);
acolor g; pd(J1);ly 10 120;keep_figure;
```




```
fit_src_struc = HN_FFCutsFits;
save(CutFitFileHN,"fit_src_struc",'-v7.3')
```